

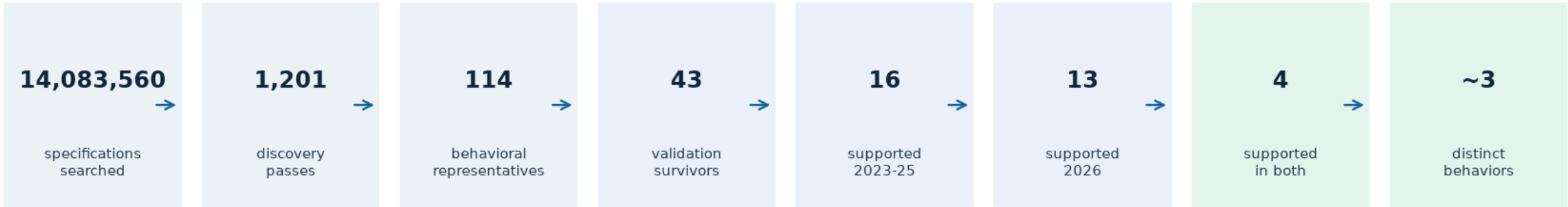
QuantLabV4

Complete research review | discovery, nulls, holdouts and prop implementation

OFFICIAL PREREGISTERED V4 RESULTS + CLEARLY MARKED POST-HOC EXTENSIONS

Research campaign: 23-25 September 2026. This review uses frozen specifications, saved JSON/NPZ trade streams, manifests and existing reports. NQ strategy points and MNQ fixed-risk dollars are kept separate.

The official search found more passing and validation-surviving strategies than its primary A/C directional nulls. Its top-five one-contract cohort passed both later aggregate tests, but the frozen \$200 MNQ prop implementation failed in 2026. The later all-43 run exposed three omitted dual-supported IDs, including the strongest 2026 and holdout net-point candidates; this is descriptive post-hoc evidence, not a revised official cohort.



OFFICIAL selection gate: 43 validation survivors -> top-five cap -> only 5 tested later. POST-HOC extension: all 43 were tested unchanged. One of the four dual-supported IDs was in the original five; three were omitted.

Executive conclusions

Question	Evidence-based answer
Search-adjusted result	Global discovery max t = 4.483, decision p = 0.0199; discovery pass count = 1,201, p = 0.0066 against A/C. The stronger path-preserving B null was weaker evidence: p = 0.171 for max t and 0.049 for pass count.
Sequential survival	43 of 114 validation representatives survived. Official five: holdout cohort +5,886 NQ points (t 1.52), 2026 +3,945 points (t 1.58). Post-hoc all 43: 16 supported in holdout, 13 in 2026, four in both.
Selection uncertainty	The top-five cap omitted three of four later dual-supported IDs. Two omitted OPEN exits share the same entries, leaving roughly two additional behaviors beyond the official dual-supported RET.
Prop suitability	Frozen \$200 MNQ sizing skipped 183 of 309 2026 signals at the one-micro floor; net -\$2,724.50 and 0/38 overlapping starts passed. A profitable one-NQ strategy is not automatically suitable for a \$2,000 evaluation loss limit.
V5 implication	Carry all qualified validation survivors forward; deduplicate behavior; design integer-MNQ deployability and evaluation geometry before freezing; strengthen path-preserving nulls and power calibration. No V5 design is created here.

Interpretation boundary: post-hoc winners, budgets and portfolio groups were discovered after later data were seen. The primary nulls remove upward drift, A and C are related, and no strategy family survived Holm adjustment. Live-forward evidence remains absent.

2. Method and evidence chain

OFFICIAL PREREGISTERED V4 RESULTS			
Stage	Dates	Purpose	Decision state
Discovery	2010-06-08 to 2018-12-31	Search, cluster and freeze rules	Observed before validation
Validation	2019-01-02 to 2022-12-30	Apply frozen rule and survival gate	43 survivors
Historical holdout	2023-01-03 to 2025-12-31	Test frozen top five	Official cohort test
Sealed 2026	2026-01-02 to 2026-08-10	Further untouched test of top five	Official forward test
All-43 extension	Same two later partitions	Run all validation survivors unchanged	Post-hoc descriptive

A strategy specification combines a trigger, filters, direction rule, RTH session window and exit. The 14.08 million-address universe was searched on 2010-18 only. Daily P&L correlation at 0.6 grouped 1,201 discovery passes into 114 representatives. Their parameters and later-data access were pinned by freezes and a stage ledger.

Execution enters on the next bar, resolves same-bar stop/target conservatively, and requires a one-tick target trade-through. Baseline NQ round-trip cost is \$14 (0.7 points), stress cost \$26 (1.3 points); NQ points x \$20 gives one-contract dollars. Features passed future-rewrite causal tests. Null markets repeat the whole search to ask whether a huge search would find apparently good rules even when the intended predictive structure is broken.

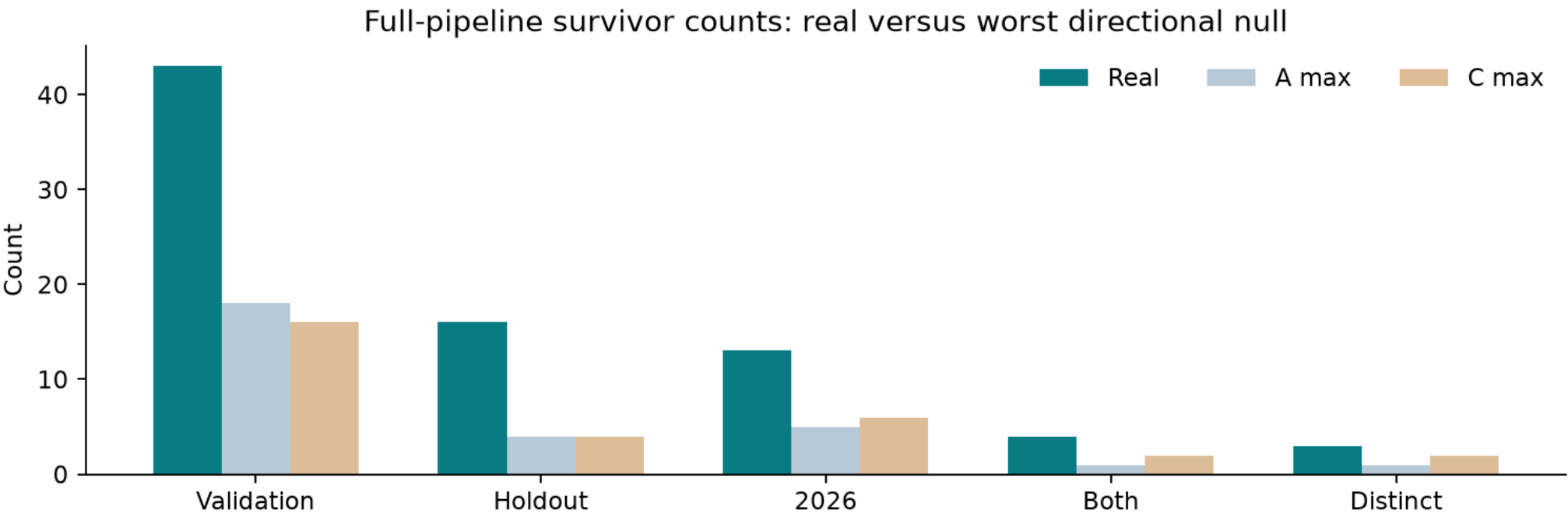
Searching millions of rules does not by itself invalidate a result: it changes the chance benchmark. The comparison must repeat selection over the same large universe. The A/C directional nulls do so; path-preserving B tests a different, tougher alternative that retains real 30-minute price paths. V4's claim is strongest for overall search yield, narrower for unique timing structure.

3. Search and null-market evidence

OFFICIAL PREREGISTERED V4 RESULTS + POST-HOC EXTENSION							
Metric	Real	A median	A max	C median	C max	p A / p C	Evidence
Discovery max eligible t	4,483	3.55	4.831	3.52	4.466	.0199 / .0066	official 150+150
Discovery passes	1,201	24	285	16	297	.0066 / .0066	official 150+150
Validation survivors	43	4	18	3	16	.0099 / .0099	post-hoc 100+100
Holdout supported	16	0	4	0	4	.0099 / .0099	post-hoc 100+100
2026 supported	13	0	5	0	6	.0099 / .0099	post-hoc 100+100
Supported both	4	0	1	0	2	.0099 / .0099	post-hoc 100+100
Distinct both	3	0	1	0	2	.0099 / .0099	post-hoc 100+100

The full all-survivor pipeline is well above the largest A or C world: real 43 / 16 / 13 / 4 / about 3 versus best A 18 / 4 / 5 / 1 / 1 and best C 16 / 4 / 6 / 2 / 2. Each extension p is a within-kind plus-one empirical tail with a 0.0099 floor. A/C are related, and their validation cohorts average only about 4.5, so later counts are not a count-matched generalization test.

Contrary evidence: Null B, which preserves realistic 30-minute paths, gives p = 0.171 for global max t and 0.0488 for passes. None of 14 family max-t findings survives Holm adjustment (minimum adjusted p 0.093). The sign-flip nulls remove market drift; earlier researcher exposure to V2/V3 also limits perfect blinding.



4. All 43 frozen validation survivors

POST-HOC ALL-43 DESCRIPTIVE TEST

Complete cross-period table. All IDs and specifications were frozen at validation; sorting below puts dual support first solely for reading. H = 2023-25; F = 2026. NQ net, drawdown and stress are points after baseline/stress cost. Classifications are the original per-strategy rule and are unchanged.

ID	Family	Rule (trigger / filters / exit)	Dir	Sessio n	Val n	Val net	Val PF	Val t	H n	H net	H PF	H t	H DD	H class	F n	F net	F PF	F t	F DD	F class	Both
744a501f	OPEN	OPEN:opening_move:X=5,thr=1.0; clvvol_aligned,jump_recent; S2_T120	cont_both	rth	277	2,505.4	1.34	1.88	208	2,260.7	1.31	1.65	777.4	SUPPORTED	50	1,443.0	1.53	1.27	701.8	SUPPORTED	YES
ec410944	OPEN	OPEN:opening_move:X=5,thr=1.0; clvvol_aligned,jump_recent; S1_T30	cont_both	rth	277	1,852.8	1.45	2.41	208	1,051.4	1.24	1.26	616.7	SUPPORTED	50	841.2	1.48	1.24	243.4	SUPPORTED	YES
941bd139	PATH	PATH:er_cross:H=15,thr=0.5; semi_aligned,vr_high; S2_R3	cont_both	rth	599	3,839.7	1.21	1.74	393	3,102.4	1.20	1.36	1,653.8	SUPPORTED	71	3,365.1	2.09	2.23	467.4	SUPPORTED	YES
40aae8a2	RET	RET:open_retz_cross:thr=2.0; semi_aligned,volreg_low; S2_R3	cont_both	rth	418	5,335.4	1.52	3.22	291	2,392.1	1.25	1.50	1,138.5	SUPPORTED	74	3,270.5	2.13	2.36	468.4	SUPPORTED	YES
6d8c3050	OPEN	OPEN:overnight_return:X=15,thr=0.5; vwap_aligned; S1_T60	cont_long	rth	128	830.9	1.43	1.63	88	715.4	1.37	1.31	557.6	SUPPORTED	28	-554.6	0.62	-1.10	696.8	FAILED	NO
158e6884	PATH	PATH:er_cross:H=15,thr=0.5; er_high,sgnvol_aligned; S2_T120	cont_long	rth	122	1,475.8	1.93	2.41	95	1,198.0	1.63	1.11	395.2	SUPPORTED	18	359.9	1.61	0.79	282.1	INCONCLUSIVE	NO
4c267443	PATH	PATH:er_cross:H=30,thr=0.5; rvol_high,semi_aligned; S2_R3	cont_long	rth	102	1,836.3	2.07	2.68	74	1,245.7	1.50	1.09	429.3	SUPPORTED	16	-969.2	0.16	-2.98	969.2	FAILED	NO
e27abc9c	PATH	PATH:er_cross:H=15,thr=0.5; volreg_low,vr_high; S2_EOD	cont_both	rth	425	3,481.3	1.35	2.30	261	-815.7	0.91	-0.58	2,089.1	FAILED	50	1,449.2	1.92	1.60	567.3	SUPPORTED	NO
f6922976	PATH	PATH:ols_t_cross:H=30,thr=3.0; open_aligned,range_top_for_long; S2_EOD	rev_long	rth	368	2,433.6	1.34	1.98	272	1,529.6	1.21	1.25	478.7	SUPPORTED	62	-356.6	0.85	-0.50	763.6	FAILED	NO
a6afc475	PERS	PERS:momentum_at_persistence_onset;; open_aligned,trend240_aligned; S1_EOD	cont_both	rth	497	2,748.1	1.27	1.85	334	2,242.0	1.24	1.33	870.9	SUPPORTED	63	873.7	1.32	0.74	713.8	INCONCLUSIVE	NO
969b7274	REOPEN	REOPEN:orb:X=30; er_low; S2_EOD	rev_both	rth	471	4,121.3	1.24	1.62	409	396.2	1.02	0.15	3,718.4	INCONCLUSIVE	82	2,126.6	1.43	1.27	1,033.6	SUPPORTED	NO
b2319ed6	REOPEN	REOPEN:orb:X=30; on_aligned,sgnvol_aligned; S1_R3	cont_long	rth	107	1,149.1	1.60	1.91	63	-141.9	0.93	-0.25	517.4	FAILED	15	1,120.0	4.22	2.12	157.6	SUPPORTED	NO
14478681	RET	RET:retz_cross:H=15,thr=2.0; sgnvol_aligned,volreg_low; S2_T30	cont_short	rth	346	2,398.3	1.72	3.45	196	1,359.5	1.46	1.86	727.0	SUPPORTED	48	268.9	1.22	0.47	348.0	INCONCLUSIVE	NO
326df064	RET	RET:trend_agree_full;; er_high,trend240_aligned; S2_R3	cont_both	rth	190	3,308.8	1.80	2.82	137	3,052.6	1.65	2.05	926.2	SUPPORTED	24	-140.5	0.90	-0.19	817.3	FAILED	NO
ae7eecdc	RET	RET:open_retz_cross:thr=1.5; on_aligned,trend60_aligned; S1_R3	cont_long	rth	188	1,603.4	1.48	2.12	137	-25.2	0.99	-0.03	565.7	FAILED	35	973.2	1.76	1.38	364.8	SUPPORTED	NO
b6a51de6	RET	RET:retz_cross:H=5,thr=1.5; range_top_for_long,semi_aligned; S2_R2	rev_long	rth	253	2,010.4	1.37	1.88	167	1,175.1	1.24	1.00	870.3	SUPPORTED	42	312.6	1.18	0.43	378.1	INCONCLUSIVE	NO
d24243de	RET	RET:open_retz_cross:thr=2.0; sgnvol_aligned,vr_high; S2_R3	cont_both	rth	340	4,511.5	1.43	2.47	215	5,273.2	1.64	2.61	670.7	SUPPORTED	53	732.4	1.26	0.63	732.6	INCONCLUSIVE	NO
1dbad4cd	TECH	TECH:cci_cross:thr=100.0; open_aligned,range_top_for_long; S2_R3	rev_both	rth	784	6,250.0	1.32	2.87	622	2,276.4	1.11	1.02	2,086.5	SUPPORTED	137	638.4	1.09	0.39	1,194.0	INCONCLUSIVE	NO
39346c8a	TECH	TECH:rsi_extreme:hi=80,lo=20; sgnvol_aligned,trend240_aligned; S2_EOD	cont_long	rth	230	3,229.3	1.64	2.40	185	1,481.2	1.23	0.90	853.9	INCONCLUSIVE	28	1,479.6	2.54	1.80	314.2	SUPPORTED	NO
efad812f	TECH	TECH:rsi_extreme:hi=80,lo=20; clvvol_aligned,volreg_low; S2_T30	cont_both	rth	347	1,957.8	1.56	2.79	227	729.9	1.24	1.14	431.8	SUPPORTED	44	165.2	1.10	0.23	615.7	INCONCLUSIVE	NO
e61bf491	VOL	VOL:obv_cross:W=120,thr=0.2; on_aligned,trend240_aligned; S2_R2	cont_long	rth	134	1,254.9	1.45	1.64	94	274.2	1.08	0.30	831.0	INCONCLUSIVE	26	1,136.3	2.05	1.42	382.7	SUPPORTED	NO
b5e4721b	VWAP	VWAP:acceptance:N=10,anchor=rth; on_aligned,sgnvol_aligned; S1_EOD	cont_long	rth	190	1,643.5	1.49	1.94	118	-51.3	0.99	-0.06	747.8	FAILED	34	1,030.2	1.74	1.10	331.1	SUPPORTED	NO
47c1bbe9	XMKT	XMKT:er_disagree:H=15,thr=0.6; first_hour; S2_T120	cont_both	rth	205	2,556.5	1.60	2.49	107	559.6	1.16	0.63	460.0	INCONCLUSIVE	23	2,448.7	3.17	1.95	418.8	SUPPORTED	NO
d24a71f3	XMKT	XMKT:pers_disagree:thr=0.4; first_hour,vr_high; S2_EOD	cont_both	rth	126	2,357.8	1.52	1.69	84	983.5	1.25	0.82	741.6	INCONCLUSIVE	15	766.0	1.98	1.00	389.9	SUPPORTED	NO

ID	Family	Rule (trigger / filters / exit)	Dir	Session	Val n	Val net	Val PF	Val t	H n	H net	H PF	H t	H DD	H class	F n	F net	F PF	F t	F DD	F class	Both
db0a9135	XMKT	XMKT:pers_disagree:thr=0.4; first_hour,volreg_low; S1_T120	cont_both	rth	173	2,274.2	1.79	2.76	86	717.5	1.35	1.12	463.6	SUPPORTED	9	323.0	1.83	0.67	153.4	INCONCLUSIVE	NO
9757cb9e	DIST	DIST:semi_cross:W=30,thr=0.7; es_confirms,trend60_aligned; S2_EOD	cont_both	rth	921	9,111.8	1.36	3.29	639	1,686.2	1.06	0.51	2,263.2	INCONCLUSIVE	131	151.8	1.02	0.08	1,708.6	INCONCLUSIVE	NO
dc5c4f13	DIST	DIST:semi_cross:W=30,thr=0.5; on_aligned,rvol_high; S2_T120	cont_both	rth	128	1,584.2	1.59	1.93	90	-128.0	0.96	-0.17	866.9	FAILED	26	-90.5	0.92	-0.17	444.8	FAILED	NO
01e36ea2	JUMP	JUMP:jump_then_smooth;; on_aligned,semi_aligned; S1_T60	cont_long	rth	96	620.5	1.51	1.56	51	-564.7	0.63	-1.40	763.1	FAILED	22	-612.7	0.43	-1.71	880.2	FAILED	NO
d3ae0d68	JUMP	JUMP:jump_then_smooth;; on_aligned,trend240_aligned; S2_T120	cont_both	rth	106	1,542.1	1.78	2.04	99	-1,004.8	0.76	-1.09	1,529.4	FAILED	30	-242.8	0.87	-0.31	1,419.0	FAILED	NO
0d7c8bc5	OPEN	OPEN:opening_move:X=30,thr=0.5; on_aligned,sgnvol_aligned; S1_T60	cont_both	rth	112	1,095.1	1.63	2.01	73	-166.1	0.92	-0.33	610.5	FAILED	21	-480.7	0.59	-0.98	820.1	FAILED	NO
d525041d	OPEN	OPEN:opening_move:X=30,thr=0.5; on_aligned,range_top_for_long; S1_T30	cont_both	rth	114	805.0	1.62	1.92	94	158.0	1.09	0.33	628.3	INCONCLUSIVE	27	-703.7	0.44	-1.67	803.0	FAILED	NO
f0e42591	OPEN	OPEN:opening_move:X=30,thr=0.5; on_aligned,trend60_aligned; S2_T120	cont_both	rth	130	1,547.5	1.52	1.77	98	334.4	1.08	0.31	966.0	INCONCLUSIVE	26	-427.7	0.76	-0.56	969.6	FAILED	NO
3315e536	REOPEN	REOPEN:orb:X=15; on_aligned,open_aligned; S2_T120	cont_long	rth	129	1,154.7	1.46	1.70	80	-844.7	0.74	-1.13	1,027.4	FAILED	26	-242.9	0.83	-0.36	930.3	FAILED	NO
653fc4da	REOPEN	REOPEN:orb:X=15; first_hour,on_aligned; S1_R2	cont_both	rth	227	2,080.1	1.36	1.99	179	987.9	1.16	0.88	759.8	INCONCLUSIVE	47	-518.4	0.82	-0.62	1,420.9	FAILED	NO
0175e8eb	RET	RET:open_retz_cross:thr=1.5; on_aligned,volreg_low; S2_T120	cont_long	rth	190	2,006.0	1.81	3.02	119	-53.3	0.98	-0.09	850.9	FAILED	33	382.6	1.52	0.90	368.0	INCONCLUSIVE	NO
8f2ddfea	RET	RET:retz_cross:H=30,thr=1.5; corr_low,open_aligned; S2_R2	cont_both	rth	92	1,434.8	1.93	2.47	17	-4.4	0.99	-0.01	548.0	FAILED	2	155.6	n/a	3.38	0.0	INCONCLUSIVE	NO
96fe2d18	RET	RET:retz_cross:H=30,thr=1.5; corr_low,range_top_for_long; S1_T60	cont_both	rth	104	696.2	1.64	1.79	19	-30.8	0.91	-0.18	240.1	FAILED	3	103.4	2.97	0.63	52.5	INCONCLUSIVE	NO
98cb83b1	RET	RET:retz_cross:H=30,thr=1.5; clvvol_aligned,corr_low; S1_T30	cont_both	rth	69	409.2	1.81	1.83	11	52.3	1.30	0.38	109.9	INCONCLUSIVE	2	21.3	2.95	0.49	10.9	INCONCLUSIVE	NO
ba1e53e9	RET	RET:open_retz_cross:thr=1.5; clvvol_aligned,on_aligned; S1_R3	cont_long	rth	146	1,342.8	1.53	2.02	99	13.7	1.01	0.02	660.9	INCONCLUSIVE	30	674.5	1.54	0.98	294.5	INCONCLUSIVE	NO
edbc44bd	RET	RET:trend_agree_full;; er_high,sgnvol_aligned; S1_T120	cont_long	rth	127	1,088.3	1.62	1.87	103	32.2	1.01	0.06	999.9	INCONCLUSIVE	14	-182.3	0.70	-0.56	339.0	FAILED	NO
60857891	VWAP	VWAP:dist_cross:anchor=glx,thr=3.0; sgnvol_aligned,volreg_low; S2_R3	cont_long	rth	317	3,305.4	1.69	3.47	199	800.2	1.16	0.88	1,225.5	INCONCLUSIVE	50	266.3	1.15	0.39	474.5	INCONCLUSIVE	NO
adf881cb	VWAP	VWAP:dist_cross:anchor=glx,thr=1.5; jump_recent,on_aligned; S1_T120	cont_long	rth	124	928.5	1.46	1.58	83	-379.9	0.84	-0.71	525.5	FAILED	26	-146.7	0.88	-0.25	405.8	FAILED	NO
c790712e	XMKT	XMKT:er_disagree:H=15,thr=0.4; corr_low,first_hour; S2_T60	cont_both	rth	147	1,804.8	1.94	2.99	47	-352.1	0.73	-0.85	530.7	FAILED	6	-11.9	0.97	-0.03	266.1	FAILED	NO

Full 28-character IDs and exact JSON specifications are in Appendix A. A short ID is shown here to keep every numeric column legible. The appendix and the source JSON disambiguate every row.

5. Four dual-supported IDs, about three behaviors

POST-HOC ALL-43 DESCRIPTIVE TEST

All four use continuation variants. The two OPEN IDs trigger on the same opening move and filters; their exits differ, and the smaller holdout stream has 208/208 matching entries. They are approximately one behavior for diversification purposes.

1. PATH | Q4-941bd1394023b222a84fb0ab

Mechanical reading: At the RTH signal, take the continuation direction when the 15-minute efficiency-ratio cross above 0.5; semivariance agrees and the volatility regime is high. Use next-bar entry, a 2x volatility stop and the frozen 3R target (or stop/EOD if first). Frozen code: PATH / er cross / H=15,thr=0.5; cont both; filters semi_aligned, vr_high; exit S2_R3; RTH. Exact feature formulas and threshold units remain defined in the V4 feature catalog and strategy engine.

Period	Trades	Net NQ pts	PF	t	Win	Exp pts	Max DD pts	Stress net pts
Discovery 2010-18	not saved here	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Validation 2019-22	599	3,839.7	1.21	1.74	n/a	6.4	n/a	3,480.3
Holdout 2023-25	393	3,102.4	1.20	1.36	44.3%	7.9	1,653.8	2,866.6
2026	71	3,365.1	2.09	2.23	56.3%	47.4	467.4	3,322.5

Trade-stream detail (NQ pts): holdout average winner 105.5, average loser -69.7, worst loss streak 19, average underwater depth 550.9, longest underwater run 102 trades, worst day -644.1, worst week -393.6; 2026 winner 161.7, loser -100.0, streak 4, average depth 155.9, longest underwater run 7 trades, worst day -252.2, worst week -216.2.

2. RET | Q4-40aae8a2c827fb779e4dd7b9

Mechanical reading: At the RTH signal, take the continuation direction when the return from the RTH open crosses a 2.0 z-score; semivariance agrees and the volatility regime is low. Use next-bar entry, a 2x volatility stop and the frozen 3R target (or stop/EOD if first). Frozen code: RET / open retz cross / thr=2.0; cont both; filters semi_aligned, volreg_low; exit S2_R3; RTH. Exact feature formulas and threshold units remain defined in the V4 feature catalog and strategy engine.

Period	Trades	Net NQ pts	PF	t	Win	Exp pts	Max DD pts	Stress net pts
Discovery 2010-18	not saved here	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Validation 2019-22	418	5,335.4	1.52	3.22	n/a	12.8	n/a	5,084.6
Holdout 2023-25	291	2,392.1	1.25	1.50	46.7%	8.2	1,138.5	2,217.5
2026	74	3,270.5	2.13	2.36	55.4%	44.2	468.4	3,226.1

Trade-stream detail (NQ pts): holdout average winner 88.1, average loser -61.9, worst loss streak 9, average underwater depth 351.1, longest underwater run 67 trades, worst day -224.1, worst week -222.8; 2026 winner 150.4, loser -87.7, streak 4, average depth 149.8, longest underwater run 10 trades, worst day -175.7, worst week -423.3.

3. OPEN | Q4-744a501fbc00d68e9d8e0ad5

Mechanical reading: At the RTH signal, take the continuation direction when the first-five-minute opening move crosses 1.0; close-location-volume agrees and a recent jump is present. Use next-bar entry, a 2x volatility stop and the frozen 120-minute time exit (or stop/EOD if first). Frozen code: OPEN / opening move / X=5,thr=1.0; cont both; filters clvvol_aligned, jump_recent; exit S2_T120; RTH. Exact feature formulas and threshold units remain defined in the V4 feature catalog and strategy engine.

Period	Trades	Net NQ pts	PF	t	Win	Exp pts	Max DD pts	Stress net pts
Discovery 2010-18	not saved here	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Validation 2019-22	277	2,505.4	1.34	1.88	n/a	9.0	n/a	2,339.2
Holdout 2023-25	208	2,260.7	1.31	1.65	51.9%	10.9	777.4	2,135.9
2026	50	1,443.0	1.53	1.27	56.0%	28.9	701.8	1,413.0

Trade-stream detail (NQ pts): holdout average winner 87.5, average loser -71.9, worst loss streak 7, average underwater depth 307.4, longest underwater run 36 trades, worst day -172.2, worst week -205.9; 2026 winner 148.8, loser -123.8, streak 4, average depth 233.5, longest underwater run 10 trades, worst day -217.4, worst week -310.1.

4. OPEN | Q4-ec4109445a1e4b4ef7b4805c

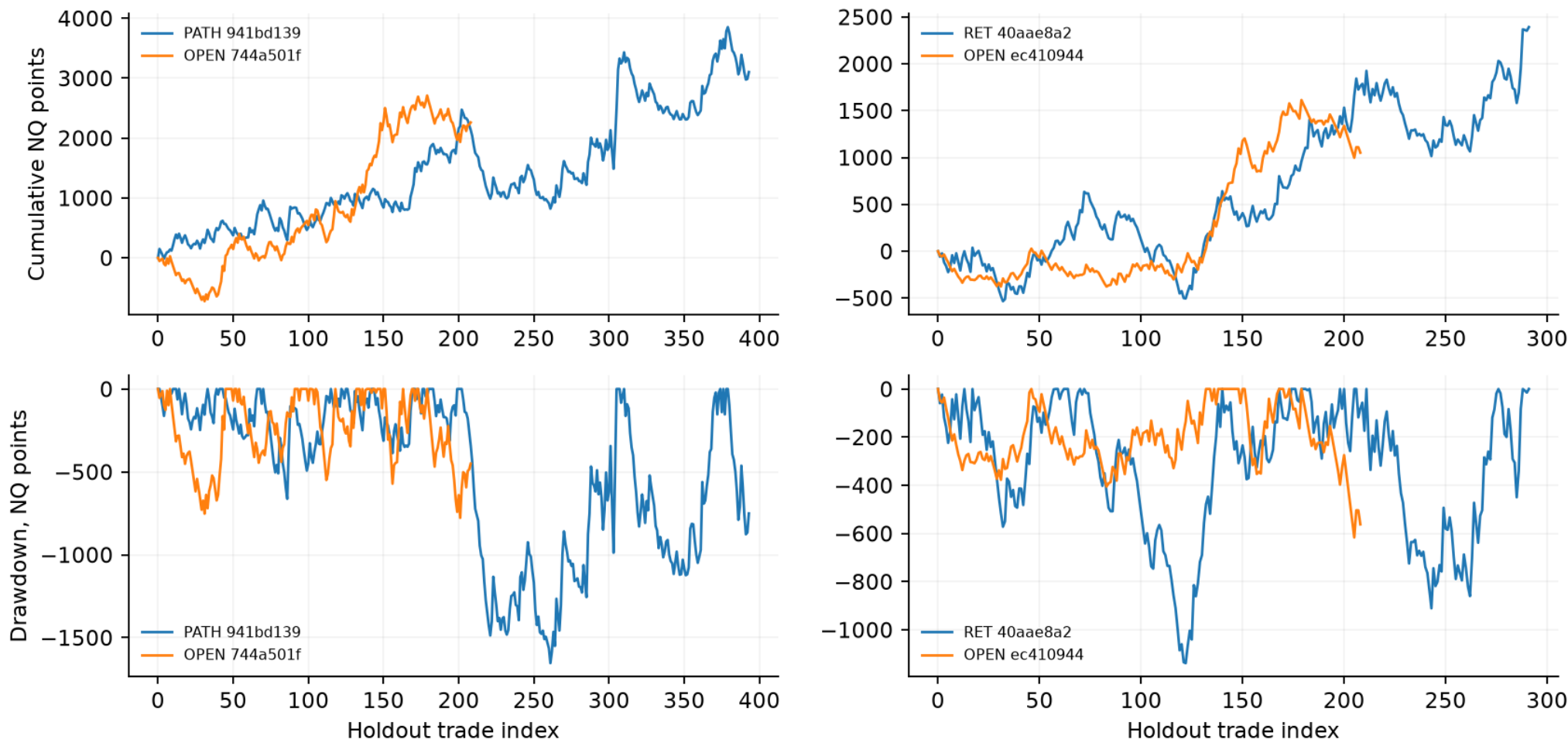
Mechanical reading: At the RTH signal, take the continuation direction when the first-five-minute opening move crosses 1.0; close-location-volume agrees and a recent jump is present. Use next-bar entry, a 1x volatility stop and the frozen 30-minute time exit (or stop/EOD if first). Frozen code: OPEN / opening move / X=5,thr=1.0; cont both; filters clvvol_aligned, jump_recent; exit S1_T30; RTH. Exact feature formulas and threshold units remain defined in the V4 feature catalog and strategy engine.

Period	Trades	Net NQ pts	PF	t	Win	Exp pts	Max DD pts	Stress net pts
Discovery 2010-18	not saved here	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Validation 2019-22	277	1,852.8	1.45	2.41	n/a	6.7	n/a	1,686.6
Holdout 2023-25	208	1,051.4	1.24	1.26	47.1%	5.1	616.7	926.6
2026	50	841.2	1.48	1.24	54.0%	16.8	243.4	811.2

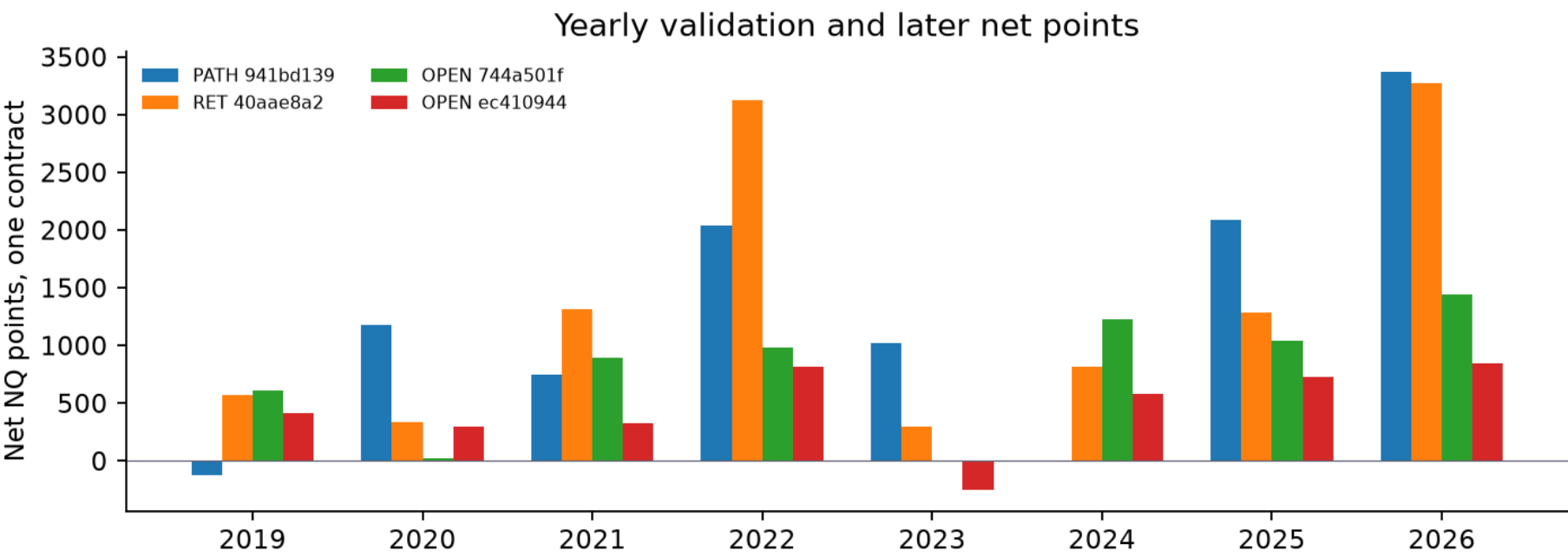
Trade-stream detail (NQ pts): holdout average winner 54.9, average loser -39.3, worst loss streak 5, average underwater depth 227.9, longest underwater run 85 trades, worst day -92.2, worst week -162.7; 2026 winner 96.2, loser -76.3, streak 3, average depth 89.2, longest underwater run 8 trades, worst day -98.7, worst week -155.9.

6. Trade paths and selection uncertainty

POST-HOC ALL-43 DESCRIPTIVE TEST



Lines are individual one-NQ trade-sequence paths over 2023-25, not the fixed-risk MNQ portfolio. A strategy can earn thousands of NQ points while its separate MNQ evaluation path breaches a \$2,000 account loss limit.



7. Other informative strategies and top-five cap

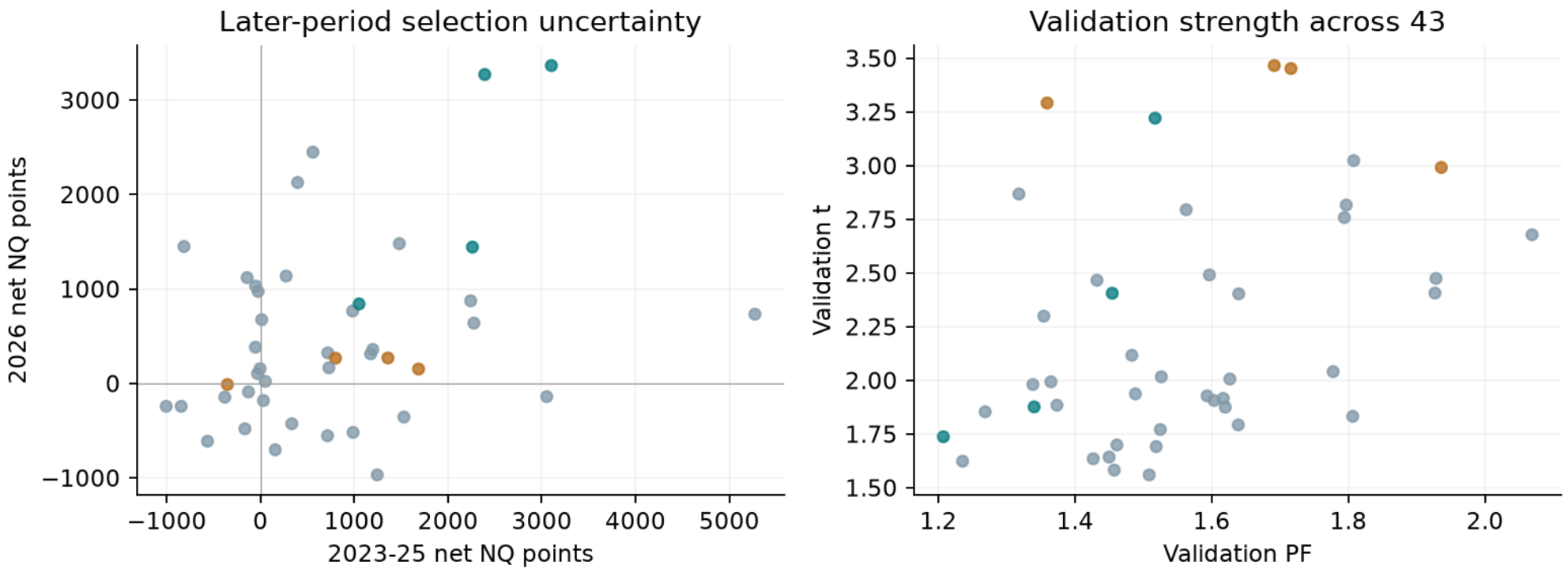
POST-HOC COMPARISON TO OFFICIAL FIVE

Highest holdout net: Q4-d24243de4d067e4f945b7e1b (RET), 5,273.2 NQ points. Highest 2026 net: Q4-941bd1394023b222a84fb0ab (PATH), 3,365.1 points. Neither belonged to the official five. Strong omitted PATH: Q4-941bd1394023b222a84fb0ab; strong omitted RET: Q4-d24243de4d067e4f945b7e1b (descriptive holdout rankings).

Official five ID	Family	Validation net	Holdout net / class	2026 net / class
Q4-60857891da97d59c25f0f86c	VWAP	3,305.4	800.2 / INCONCLUSIVE	266.3 / INCONCLUSIVE
Q4-9757cb9ee92c11b68d2ad60b	DIST	9,111.8	1,686.2 / INCONCLUSIVE	151.8 / INCONCLUSIVE
Q4-14478681d690cde198df1979	RET	2,398.3	1,359.5 / SUPPORTED	268.9 / INCONCLUSIVE
Q4-40aae8a2c827fb779e4dd7b9	RET	5,335.4	2,392.1 / SUPPORTED	3,270.5 / SUPPORTED
Q4-c790712e4f51f6f4268b0033	XMKT	1,804.8	-352.1 / FAILED	-11.9 / FAILED

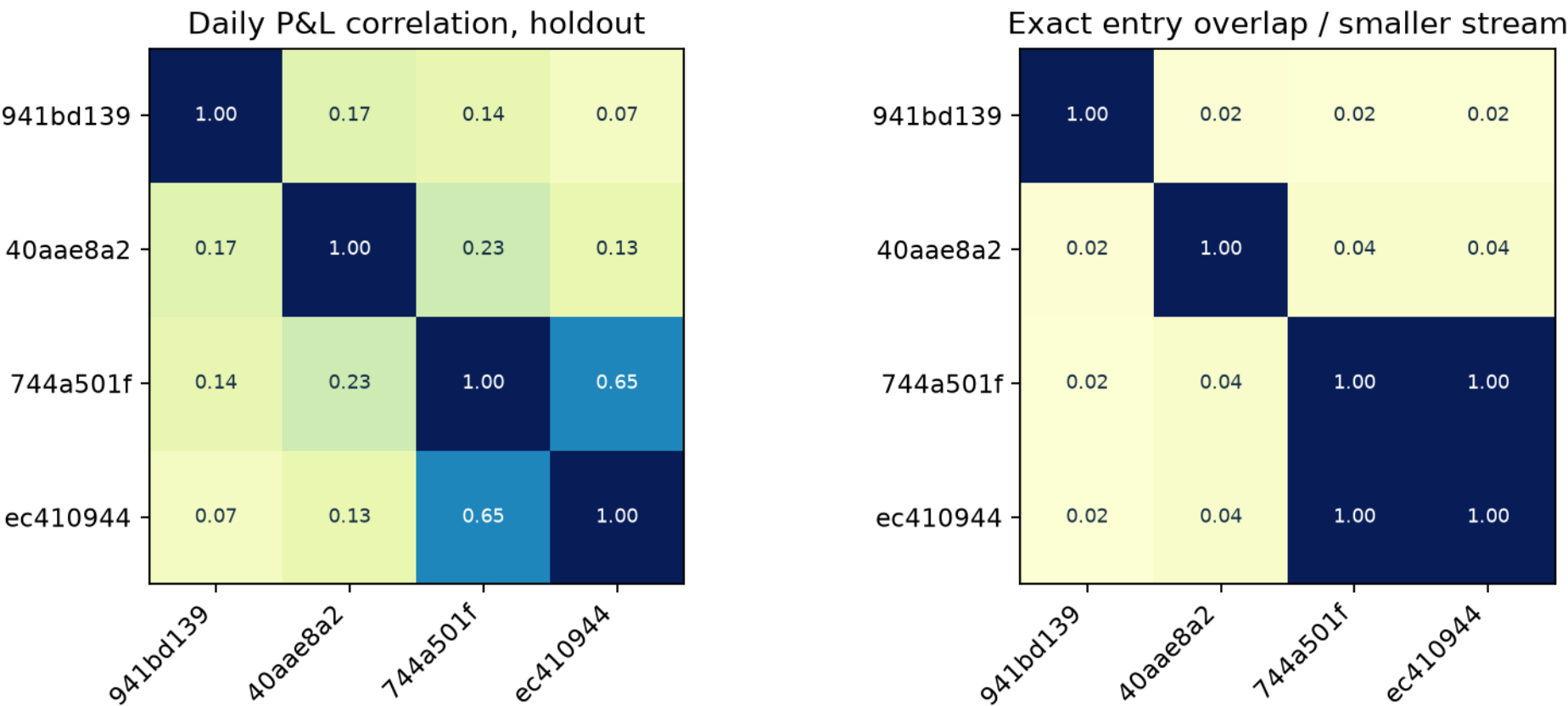
Original five -> 1 dual-supported ID. All 43 -> 4 dual-supported IDs, with the omitted three representing roughly two additional behaviors. The cap appears overly restrictive as a research design choice, but identifying its misses required later outcomes; these omissions are not prospective selection evidence.

XMKT efficiency disagreement failed in the official holdout and 2026 tests. Other high-validation survivors also collapsed, showing that validation strength did not reliably rank later outcomes. The full table preserves all failures and inconclusive IDs alongside winners.



8. Correlation, entry overlap and concentration

POST-HOC ALL-43 DESCRIPTIVE TEST



Pair	Daily corr H / 2026	Same entry H / 2026	Same direction H / 2026	Simultaneous fraction H / 2026
941bd139 x 40aae8a2	0.17 / 0.22	6 / 0	6 / 0	37.5% / 39.4%
941bd139 x 744a501f	0.14 / 0.24	4 / 2	4 / 2	33.2% / 26.0%
941bd139 x ec410944	0.07 / 0.28	4 / 2	4 / 2	24.0% / 20.0%
40aae8a2 x 744a501f	0.23 / 0.35	9 / 1	9 / 1	36.1% / 48.0%
40aae8a2 x ec410944	0.13 / 0.31	9 / 1	9 / 1	28.8% / 30.0%
744a501f x ec410944	0.65 / 0.65	208 / 50	208 / 50	99.0% / 100.0%

The two OPEN variants have daily correlation 0.65 in both periods, 208 shared holdout entries and 50 shared 2026 entries, with nearly complete simultaneous overlap. PATH and RET have lower daily correlations and only a handful of exact shared entries, but 24-49% of smaller-stream trades overlap in time with another dual-supported stream. Low daily correlation is not independence: all four favor continuation, and two of 21 positive-both strategies form another descriptive TECH/PATH copy pair.

9. Drawdown and fixed-risk MNQ sensitivity

POST-HOC RISK-BUDGET DIAGNOSTICS

The strategy curves above measure one NQ contract in index points. The following figures size MNQ contracts at a dollar risk budget, apply integer sizing, a simultaneous open-risk cap of 2x budget, and stop-after-\$300 daily loss. They are different P&L processes.

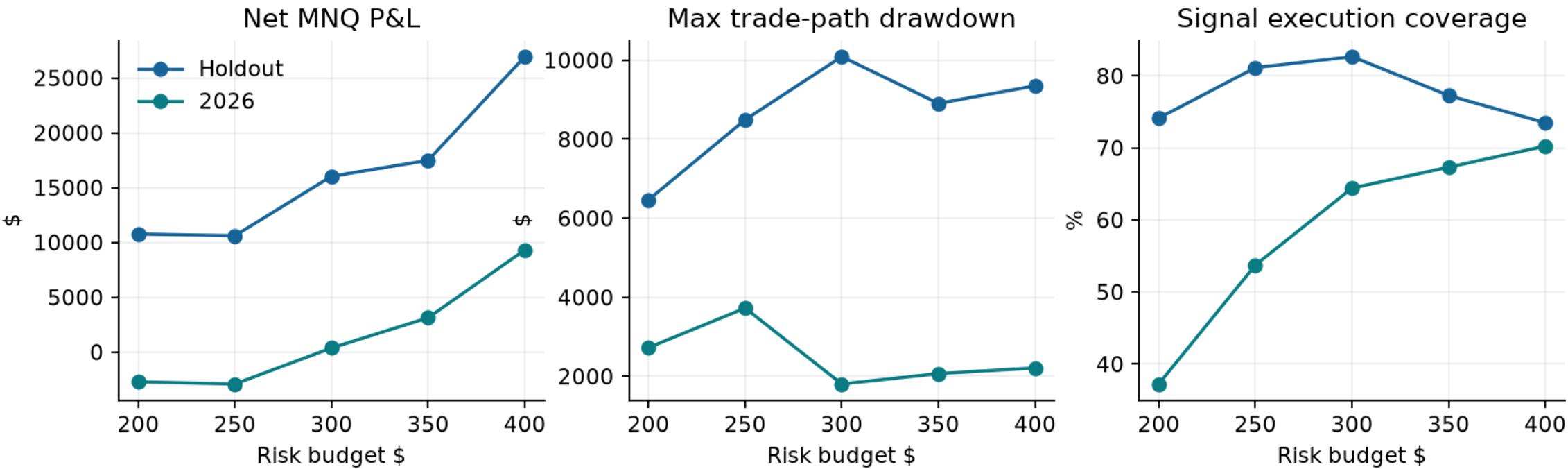
2023-25 original five

Budget	Signals	Trades	Skipped	Coverage	1-MNQ skips	Open-risk skips	Daily skips	Net MNQ \$	Max DD \$	Max trade risk \$
\$200	1372	1017	355	74.1%	240	71	44	\$10,761	\$6,464	n/a
\$250	1372	1113	259	81.1%	108	80	71	\$10,606	\$8,490	n/a
\$300	1372	1134	238	82.7%	43	91	104	\$16,026	\$10,087	n/a
\$350	1372	1060	312	77.3%	23	98	191	\$17,480	\$8,904	n/a
\$400	1372	1008	364	73.5%	7	104	253	\$26,939	\$9,348	\$400

2026 original five

Budget	Signals	Trades	Skipped	Coverage	1-MNQ skips	Open-risk skips	Daily skips	Net MNQ \$	Max DD \$	Max trade risk \$
\$200	309	115	194	37.2%	183	6	5	\$-2,724	\$2,724	n/a
\$250	309	166	143	53.7%	119	12	12	\$-2,920	\$3,724	n/a
\$300	309	199	110	64.4%	77	21	12	\$363	\$1,805	n/a
\$350	309	208	101	67.3%	44	29	28	\$3,120	\$2,068	n/a
\$400	309	217	92	70.2%	19	31	42	\$9,300	\$2,210	\$398

At \$200, 183/309 2026 signals fail even the one-MNQ floor, including 50/74 from the official RET open-continuation winner. Coverage of underlying signals was distorted. As the budget rises, more signals fit one micro, but portfolio open risk and the daily policy can still reduce executed trades. The \$400 result is +\$9,300.50 with a \$2,210 trade-sequence drawdown; neither directly equals an evaluation pass probability.



Uncapped diagnostic: the 19 signals rejected by the \$400 one-micro floor in 2026, valued at one MNQ before portfolio filtering, lost \$1,156.50 together (9/19 winners; -\$60.87 average). The uncapped portfolio earned \$6,712 versus \$9,300.50 at \$400 because admitting signals changes cap and daily exclusions. Removing the floor did not improve this observed path.

10. Generic prop-evaluation geometry

POST-HOC AND OFFICIAL RESULTS DISTINGUISHED

Generic model: \$50k account, \$3,000 profit target, \$2,000 maximum loss, EOD or intraday trailing variants, \$1,000 daily-loss rule, 50% consistency rule, stop-after-\$300-day trading policy and open-risk cap 2x per-trade budget. These are research profiles, not verified firm rules. Rolling starts use 120-session windows; 656 holdout starts and only 38 2026 starts overlap heavily.

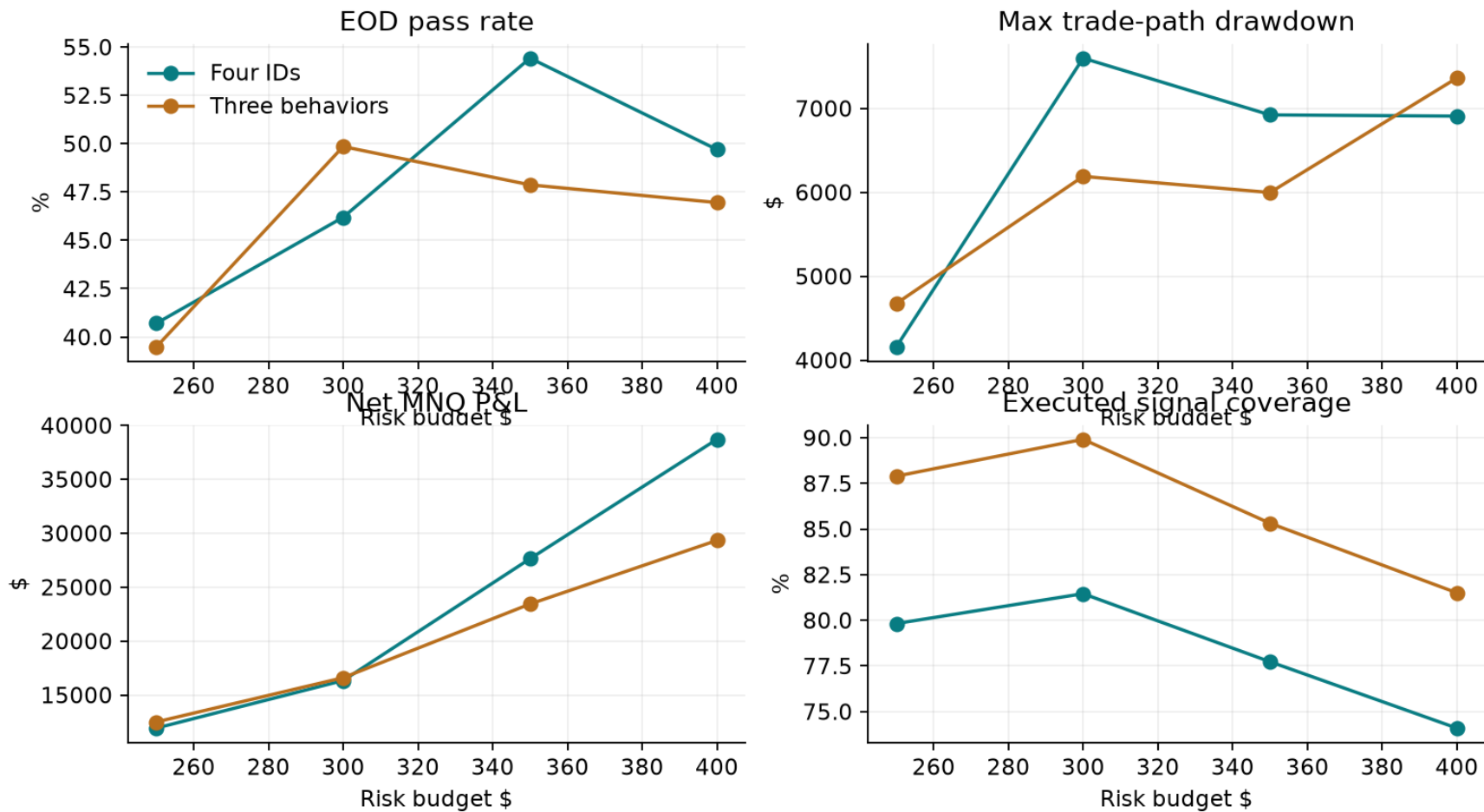
OFFICIAL frozen \$200 five-strategy result: holdout 45% pass / 49% fail and +\$10,761 MNQ net; 2026 0/38 passes, 71% fails and -\$2,724.50. Later budget variations are post-hoc.

Portfolio	Period	Budget	Starts	Pass n / %	Fail n / %	Open %	Median days	Net MNQ \$	Max DD \$	Max open risk \$	Coverage
Four dual-supported IDs	2023-25	\$250	656	267 / 40.7%	388 / 59.1%	0.2%	55.0	\$11,921	\$4,157	\$499	79.8%
Four dual-supported IDs	2023-25	\$300	656	303 / 46.2%	348 / 53.0%	0.8%	39.0	\$16,333	\$7,598	\$599	81.5%
Four dual-supported IDs	2023-25	\$350	656	357 / 54.4%	294 / 44.8%	0.8%	33.0	\$27,630	\$6,922	\$698	77.7%
Four dual-supported IDs	2023-25	\$400	656	326 / 49.7%	330 / 50.3%	0.0%	25.5	\$38,702	\$6,908	\$798	74.1%
Four dual-supported IDs	2026	\$250	38	9 / 23.7%	0 / 0.0%	76.3%	91.0	\$3,787	\$1,692	\$484	52.2%
Four dual-supported IDs	2026	\$300	38	38 / 100.0%	0 / 0.0%	0.0%	47.5	\$8,218	\$1,594	\$584	64.1%
Four dual-supported IDs	2026	\$350	38	38 / 100.0%	0 / 0.0%	0.0%	28.5	\$13,114	\$1,689	\$690	68.6%
Four dual-supported IDs	2026	\$400	38	38 / 100.0%	0 / 0.0%	0.0%	26.5	\$16,610	\$2,534	\$799	73.5%
Three behaviors	2023-25	\$250	656	259 / 39.5%	369 / 56.2%	4.3%	59.0	\$12,492	\$4,674	\$498	87.9%
Three behaviors	2023-25	\$300	656	327 / 49.8%	324 / 49.4%	0.8%	48.0	\$16,596	\$6,190	\$587	89.9%
Three behaviors	2023-25	\$350	656	314 / 47.9%	341 / 52.0%	0.2%	36.0	\$23,431	\$5,997	\$700	85.3%
Three behaviors	2023-25	\$400	656	308 / 47.0%	348 / 53.0%	0.0%	32.5	\$29,326	\$7,362	\$781	81.5%
Three behaviors	2026	\$250	38	25 / 65.8%	0 / 0.0%	34.2%	97.0	\$4,343	\$1,113	\$446	63.1%
Three behaviors	2026	\$300	38	38 / 100.0%	0 / 0.0%	0.0%	47.5	\$9,797	\$1,144	\$578	74.4%
Three behaviors	2026	\$350	38	38 / 100.0%	0 / 0.0%	0.0%	39.0	\$11,810	\$1,334	\$681	75.4%
Three behaviors	2026	\$400	38	38 / 100.0%	0 / 0.0%	0.0%	39.5	\$15,794	\$2,072	\$754	79.0%
21 positive-both IDs	2023-25	\$250	656	200 / 30.5%	456 / 69.5%	0.0%	30.0	\$14,522	\$7,772	\$500	42.1%
21 positive-both IDs	2023-25	\$300	656	221 / 33.7%	432 / 65.9%	0.5%	24.0	\$34,211	\$8,191	\$600	41.0%
21 positive-both IDs	2023-25	\$350	656	220 / 33.5%	431 / 65.7%	0.8%	23.0	\$40,178	\$9,472	\$700	36.9%
21 positive-both IDs	2023-25	\$400	656	219 / 33.4%	437 / 66.6%	0.0%	26.0	\$47,226	\$8,230	\$798	33.7%
21 positive-both IDs	2026	\$250	38	0 / 0.0%	38 / 100.0%	0.0%	n/a	\$7,142	\$2,998	\$498	31.7%
21 positive-both IDs	2026	\$300	38	26 / 68.4%	12 / 31.6%	0.0%	25.5	\$12,390	\$2,743	\$600	37.2%
21 positive-both IDs	2026	\$350	38	19 / 50.0%	19 / 50.0%	0.0%	52.0	\$9,935	\$2,838	\$700	34.6%
21 positive-both IDs	2026	\$400	38	23 / 60.5%	15 / 39.5%	0.0%	45.0	\$21,120	\$2,922	\$794	34.1%
One per family	2023-25	\$250	656	224 / 34.1%	422 / 64.3%	1.5%	36.0	\$18,126	\$10,658	\$499	62.9%
One per family	2023-25	\$300	656	130 / 19.8%	515 / 78.5%	1.7%	27.0	\$24,216	\$11,448	\$597	62.5%
One per family	2023-25	\$350	656	102 / 15.5%	554 / 84.5%	0.0%	27.5	\$12,721	\$12,709	\$698	56.4%
One per family	2023-25	\$400	656	124 / 18.9%	532 / 81.1%	0.0%	24.0	\$28,594	\$14,057	\$798	51.4%
One per family	2026	\$250	38	23 / 60.5%	13 / 34.2%	5.3%	68.0	\$5,937	\$2,846	\$494	47.7%

Portfolio	Period	Budget	Starts	Pass n / %	Fail n / %	Open %	Median days	Net MNQ \$	Max DD \$	Max open risk \$	Coverage
One per family	2026	\$300	38	7 / 18.4%	31 / 81.6%	0.0%	38.0	\$11,169	\$2,755	\$599	53.1%
One per family	2026	\$350	38	14 / 36.8%	24 / 63.2%	0.0%	33.0	\$8,036	\$3,658	\$698	51.0%
One per family	2026	\$400	38	11 / 28.9%	27 / 71.1%	0.0%	41.0	\$17,533	\$3,156	\$800	50.3%

Four dual-supported IDs at \$350: 54.4% pass, 44.8% fail, \$27,630 holdout net and \$6,922 max trade-path drawdown. Three-behavior combination at \$300: 49.8% pass, 49.4% fail, \$16,596 net and \$6,190 drawdown. Both drawdowns greatly exceed the \$2,000 evaluation loss limit; rolling starts may reset and pass before the whole-period path's maximum drawdown.

The 2026 rolling-start results are in the table, but 38/38 or 36/38 overlapping starts must never be interpreted as a 100% or 94.7% probability for a new future evaluation. The saved portfolio output does not report 90th-percentile days to pass or worst week; those metrics are intentionally omitted.



11. Descriptive portfolios and V5 lessons

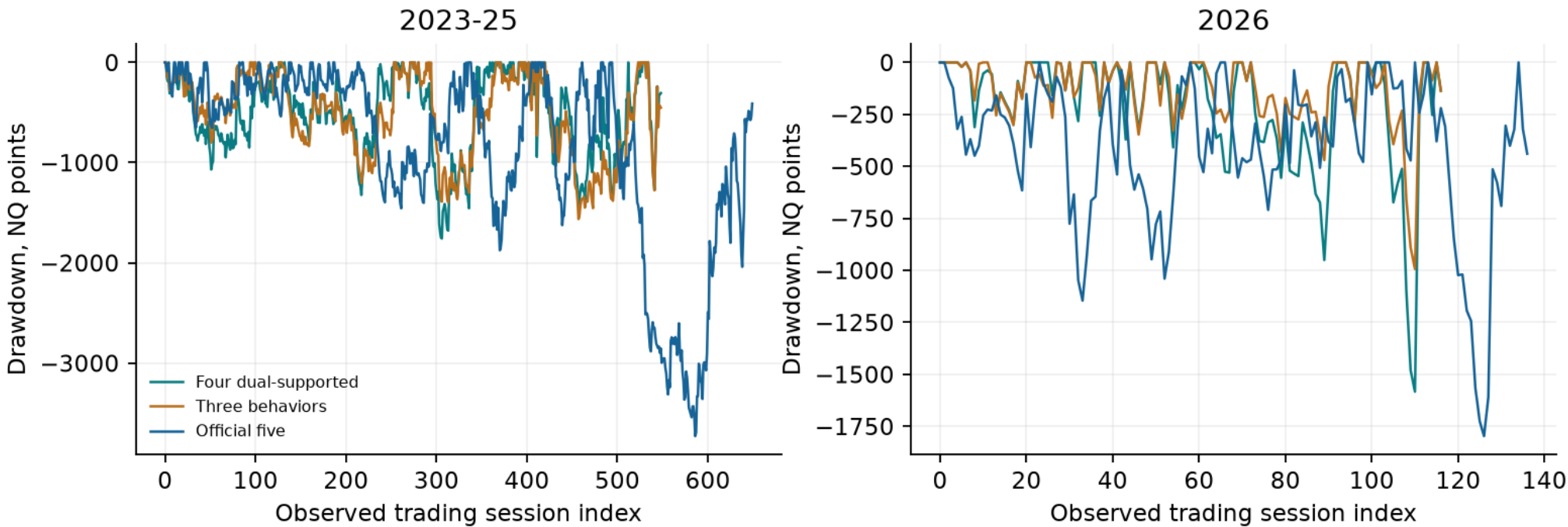
POST-HOC PORTFOLIOS; OFFICIAL FIVE KEPT SEPARATE

The table below combines stored one-NQ trade streams mechanically, with one contract per ID and no portfolio execution cap. It is a diagnostic in NQ points, not an MNQ deployable P&L. Distinct counts use the all-43 descriptive copy screen (pooled daily correlation ≥ 0.60 or at least 50% shared holdout entries); no independence claim follows.

Group	Period	IDs / beh.	Net NQ pts	PF	Max DD pts	Worst day pts	Mean pair corr	Max open IDs	Largest net share
Official five	2023-25	5 / ~5	5,885.9	1.13	3,729.0	-495.9	0.10	4	40.6%
Official five	2026	5 / ~5	3,945.5	1.27	1,797.2	-513.2	0.12	4	82.9%
Four dual-supported	2023-25	4 / ~3	8,806.5	1.24	1,757.1	-644.1	0.23	4	35.2%
Four dual-supported	2026	4 / ~3	8,919.8	1.85	1,583.6	-584.5	0.34	4	37.7%
Three behaviors	2023-25	3 / ~3	6,545.9	1.22	1,563.4	-644.1	0.12	3	47.4%
Three behaviors	2026	3 / ~3	7,476.8	1.96	993.9	-426.3	0.26	3	45.0%
Holdout-supported	2023-25	16 / ~15	30,321.4	1.29	4,932.4	-948.9	0.16	12	17.4%
Holdout-supported	2026	16 / ~15	10,572.7	1.30	3,902.9	-1,137.3	0.16	11	31.8%
2026-supported	2023-25	13 / ~11	11,467.2	1.13	3,860.6	-1,355.5	0.12	11	27.1%
2026-supported	2026	13 / ~11	21,449.7	1.90	2,487.4	-1,537.5	0.21	11	15.7%
Positive both	2023-25	21 / ~19	30,025.2	1.19	7,351.0	-1,196.9	0.12	15	17.6%
Positive both	2026	21 / ~19	21,664.8	1.44	4,441.7	-1,675.7	0.18	13	15.5%

Largest net share can exceed 100% when other members lose money. Mean pair correlation is a descriptive average of zero-filled daily streams and hides tail dependence. Max open IDs counts overlapping positions, not dollar risk. Section 10 reports actual fixed-risk MNQ combinations where the saved engine calculated them.

Drawdown calculation note: the official cohort file reports a 2023-25 maximum drawdown of 2,263.1 NQ points, while direct aggregation of the saved all-43 daily trade cashflows gives 3,729.0 points for those same five IDs. Net points and worst day reconcile. These different drawdown figures are preserved and labelled by calculation; the larger reconstructed daily drawdown is more conservative for portfolio risk review.



Portfolio P&L depends on shared entries, simultaneously open stops, integer-MNQ sizing and daily exclusions. The four-ID set double-counts the OPEN opening signal with different exits. Simple one-NQ addition is not a deployable portfolio; use the saved portfolio engine's executed MNQ results in Section 10.

V5 design lessons

- 1. Advance every qualifying validation survivor to later testing; preserve the original full set and all failures.
- 2. Separate behavior from ID count. Keep cluster-level evidence, same-entry overlap and simultaneous exposure alongside daily correlation.
- 3. Include integer-MNQ sizing, one-micro floor coverage, open-risk interaction, and an absolute \$2,000 evaluation suitability bar before freezing a prop design.
- 4. Explore approximately \$250-\$400 as a research range, then freeze a choice before new forward data; observed best rows are not a live rule.
- 5. Use stronger structure-preserving market nulls and improve power calibration. Distinguish a rejected hypothesis from an underpowered test.
- 6. Use efficient multiple-testing methods only where their assumptions hold; retain full market-level null runs as robustness checks.
- 7. Measure pass, fail, unresolved, time to pass, worst path and signal coverage, not only total P&L. Collect truly new live-forward evidence.

A. All 43 exact frozen specifications

POST-HOC APPENDIX; SPECIFICATIONS FROZEN BEFORE LATER TESTS		
Full candidate ID	Family	Frozen trigger / variant / filters / exit / session
Q4-0d7c8bc571764e5378bc4112	OPEN	OPEN / opening move / X=30,thr=0.5; cont both; filters on_aligned, sgnvol_aligned; exit S1_T60; RTH
Q4-941bd1394023b222a84fb0ab	PATH	PATH / er cross / H=15,thr=0.5; cont both; filters semi_aligned, vr_high; exit S2_R3; RTH
Q4-8f2ddfeac6111992489eb2e1	RET	RET / retz cross / H=30,thr=1.5; cont both; filters corr_low, open_aligned; exit S2_R2; RTH
Q4-40aae8a2c827fb779e4dd7b9	RET	RET / open retz cross / thr=2.0; cont both; filters semi_aligned, volreg_low; exit S2_R3; RTH
Q4-e27abc9c40a95808b286aaf8	PATH	PATH / er cross / H=15,thr=0.5; cont both; filters volreg_low, vr_high; exit S2_EOD; RTH
Q4-326df064b2c927151d6fc24c	RET	RET / trend agree full / ; cont both; filters er_high, trend240_aligned; exit S2_R3; RTH
Q4-96fe2d187f77585a991b6695	RET	RET / retz cross / H=30,thr=1.5; cont both; filters corr_low, range_top_for_long; exit S1_T60; RTH
Q4-ae7eecd86608a558045b27d	RET	RET / open retz cross / thr=1.5; cont long; filters on_aligned, trend60_aligned; exit S1_R3; RTH
Q4-a6afc475d401ac496658b786	PERS	PERS / momentum at persistence onset / ; cont both; filters open_aligned, trend240_aligned; exit S1_EOD; RTH
Q4-744a501fbc00d68e9d8e0ad5	OPEN	OPEN / opening move / X=5,thr=1.0; cont both; filters clvvol_aligned, jump_recent; exit S2_T120; RTH
Q4-d3ae0d688762e49eeec25e7a	JUMP	JUMP / jump then smooth / ; cont both; filters on_aligned, trend240_aligned; exit S2_T120; RTH
Q4-db0a91353062ca891caf8eda	XMKT	XMKT / pers disagree / thr=0.4; cont both; filters first_hour, volreg_low; exit S1_T120; RTH
Q4-d24a71f3f98b73539e298d5d	XMKT	XMKT / pers disagree / thr=0.4; cont both; filters first_hour, vr_high; exit S2_EOD; RTH
Q4-ba1e53e93129e12bb7d777b0	RET	RET / open retz cross / thr=1.5; cont long; filters clvvol_aligned, on_aligned; exit S1_R3; RTH
Q4-01e36ea2e3f5fba896554d38	JUMP	JUMP / jump then smooth / ; cont long; filters on_aligned, semi_aligned; exit S1_T60; RTH
Q4-6d8c3050c62a8e590ae96486	OPEN	OPEN / overnight return / X=15,thr=0.5; cont long; filters vwap_aligned; exit S1_T60; RTH
Q4-f6922976879d5e719f7e99c8	PATH	PATH / ols t cross / H=30,thr=3.0; rev long; filters open_aligned, range_top_for_long; exit S2_EOD; RTH
Q4-c790712e4f51f6f4268b0033	XMKT	XMKT / er disagree / H=15,thr=0.4; cont both; filters corr_low, first_hour; exit S2_T60; RTH
Q4-f0e425918ed9c2e32b39322b	OPEN	OPEN / opening move / X=30,thr=0.5; cont both; filters on_aligned, trend60_aligned; exit S2_T120; RTH
Q4-edbc44bde03d4bba60783c4b	RET	RET / trend agree full / ; cont long; filters er_high, sgnvol_aligned; exit S1_T120; RTH
Q4-b5e4721bee9e34f41277388e	VWAP	VWAP / acceptance / N=10,anchor=rth; cont long; filters on_aligned, sgnvol_aligned; exit S1_EOD; RTH
Q4-dc5c4f1370da762131955beb	DIST	DIST / semi cross / W=30,thr=0.5; cont both; filters on_aligned, rvol_high; exit S2_T120; RTH
Q4-39346c8ab50d60a228f53624	TECH	TECH / rsi extreme / hi=80,lo=20; cont long; filters sgnvol_aligned, trend240_aligned; exit S2_EOD; RTH
Q4-e61bf49122da50003c554a54	VOL	VOL / obv cross / W=120,thr=0.2; cont long; filters on_aligned, trend240_aligned; exit S2_R2; RTH
Q4-3315e536ae22ccdb042483ab	REOPEN	REOPEN / orb / X=15; cont long; filters on_aligned, open_aligned; exit S2_T120; RTH
Q4-98cb83b1751ac82049ea446c	RET	RET / retz cross / H=30,thr=1.5; cont both; filters clvvol_aligned, corr_low; exit S1_T30; RTH
Q4-47c1bbe9fc1427e364d86bf5	XMKT	XMKT / er disagree / H=15,thr=0.6; cont both; filters first_hour; exit S2_T120; RTH
Q4-9757cb9ee92c11b68d2ad60b	DIST	DIST / semi cross / W=30,thr=0.7; cont both; filters es_confirms, trend60_aligned; exit S2_EOD; RTH
Q4-1dbad4cded1e659fd193e15c	TECH	TECH / cci cross / thr=100.0; rev both; filters open_aligned, range_top_for_long; exit S2_R3; RTH
Q4-b6a51de64a0447c802114a4a	RET	RET / retz cross / H=5,thr=1.5; rev long; filters range_top_for_long, semi_aligned; exit S2_R2; RTH
Q4-ec4109445a1e4b4ef7b4805c	OPEN	OPEN / opening move / X=5,thr=1.0; cont both; filters clvvol_aligned, jump_recent; exit S1_T30; RTH
Q4-4c2674436a7c2e911ab1ce4c	PATH	PATH / er cross / H=30,thr=0.5; cont long; filters rvol_high, semi_aligned; exit S2_R3; RTH

Full candidate ID	Family	Frozen trigger / variant / filters / exit / session
Q4-14478681d690cde198df1979	RET	RET / retz cross / H=15,thr=2.0; cont short; filters sgnvol_aligned, volreg_low; exit S2_T30; RTH
Q4-653fc4dadbcc80f6ed658dc3	REOPEN	REOPEN / orb / X=15; cont both; filters first_hour, on_aligned; exit S1_R2; RTH
Q4-158e68845ba80d9490e4b658	PATH	PATH / er cross / H=15,thr=0.5; cont long; filters er_high, sgnvol_aligned; exit S2_T120; RTH
Q4-d525041d0b7efeb68425a975	OPEN	OPEN / opening move / X=30,thr=0.5; cont both; filters on_aligned, range_top_for_long; exit S1_T30; RTH
Q4-0175e8eb4fea121c87655234	RET	RET / open retz cross / thr=1.5; cont long; filters on_aligned, volreg_low; exit S2_T120; RTH
Q4-d24243de4d067e4f945b7e1b	RET	RET / open retz cross / thr=2.0; cont both; filters sgnvol_aligned, vr_high; exit S2_R3; RTH
Q4-969b7274b473016b12b50cda	REOPEN	REOPEN / orb / X=30; rev both; filters er_low; exit S2_EOD; RTH
Q4-60857891da97d59c25f0f86c	VWAP	VWAP / dist cross / anchor=glx,thr=3.0; cont long; filters sgnvol_aligned, volreg_low; exit S2_R3; RTH
Q4-efad812f7fec4376c24b34f6	TECH	TECH / rsi extreme / hi=80,lo=20; cont both; filters clvvol_aligned, volreg_low; exit S2_T30; RTH
Q4-b2319ed6a27466f8ef78245e	REOPEN	REOPEN / orb / X=30; cont long; filters on_aligned, sgnvol_aligned; exit S1_R3; RTH
Q4-adf881cbaba224976c11c316	VWAP	VWAP / dist cross / anchor=glx,thr=1.5; cont long; filters jump_recent, on_aligned; exit S1_T120; RTH

B. Full all-43 performance reference

POST-HOC APPENDIX

Section 4 contains the complete 43-row performance table with validation, holdout and 2026 figures and classifications. The machine-readable file V4_POSTHOC_ALL_VALIDATION_SURVIVORS.json holds all full-precision fields and exact classifications. The table's short IDs map to the full IDs in Appendix A.

C. Correlation matrices and pair details

POST-HOC APPENDIX

Section 8 plots the four-ID daily P&L and entry-overlap matrices. The source JSON holds all 210 pairs among 21 positive-both IDs, including holdout/2026 daily correlation, exact/same-direction entries and simultaneous fractions.

Pair among four dual-supported IDs	Holdout corr	2026 corr	Holdout entry fraction	2026 entry fraction
941bd139 x 40aae8a2	0.165	0.216	2.1%	0.0%
941bd139 x 744a501f	0.137	0.237	1.9%	4.0%
941bd139 x ec410944	0.071	0.276	1.9%	4.0%
40aae8a2 x 744a501f	0.230	0.355	4.3%	2.0%
40aae8a2 x ec410944	0.135	0.307	4.3%	2.0%
744a501f x ec410944	0.650	0.650	100.0%	100.0%

D. Fixed-risk tables and cap diagnostics

POST-HOC APPENDIX

Sections 9-10 reproduce all five \$200-\$400 original-five risk rows and all four \$250-\$400 post-hoc portfolio combinations for both later periods. Saved JSON contains per-strategy rows, EOD/intraday profiles, exact skips and further \$500-\$800/uncapped cap diagnostics. The evaluated profile should not be read as a firm's contract.

Selected \$400 cap diagnostic (original five): holdout 1,008/1,372 executed, \$26,939 net, \$9,348 maximum trade-sequence drawdown; 2026 217/309 executed, \$9,300.50 net, \$2,210 drawdown. At 2026 \$400, skip decomposition is 19 one-micro, 31 open-risk and 42 daily-policy skips; maximum actual executed single-trade stop risk is \$398.

Extended cap diagnostic, original five

Period	Cap	Signals	Executed	1-MNQ skip	Open skip	Daily skip	Net \$	Max DD \$	Max actual risk \$	EOD pass / fail
2023-25	400	1372	1008	7	104	253	\$26,939	\$9,348	\$400	43.3% / 56.7%
2023-25	500	1372	942	2	116	312	\$35,386	\$11,622	\$500	37.5% / 62.5%
2023-25	600	1372	920	0	119	333	\$52,016	\$12,704	\$600	26.8% / 73.2%
2023-25	800	1372	914	0	120	338	\$71,296	\$17,462	\$800	20.1% / 79.6%
2023-25	UNCAPPED	1372	1012	0	106	254	\$26,716	\$9,582	\$590	41.6% / 57.9%
2026	400	309	217	19	31	42	\$9,300	\$2,210	\$398	13.2% / 86.8%
2026	500	309	202	6	37	64	\$9,867	\$3,540	\$500	21.1% / 78.9%
2026	600	309	202	0	36	71	\$12,446	\$4,193	\$598	21.1% / 78.9%
2026	800	309	191	0	40	78	\$22,788	\$5,550	\$794	18.4% / 81.6%
2026	UNCAPPED	309	217	0	39	53	\$6,712	\$3,863	\$576	13.2% / 86.8%

D2. Full saved post-hoc per-strategy risk rows

POST-HOC APPENDIX

Each row is an individual strategy traded alone under the saved fixed-risk policy. This does not sum to a portfolio because open-risk and daily rules interact across strategies. Budget \$200 was studied for the original five only; the all-43 extension's saved 21-positive strategy diagnostic covers \$250-\$400.

2023-25 holdout

ID	Fam	Budget	Signals	Exec	Coverage	1-MNQ	Open	Daily	Net MNQ \$	Max DD \$	Max risk \$	Pass	Fail
941bd139	PATH	\$250	393	342	87.0%	50	0	1	\$2,550	\$3,866	\$250	9.0%	35.4%
941bd139	PATH	\$300	393	363	92.4%	26	0	4	\$4,465	\$3,863	\$300	15.9%	39.5%
941bd139	PATH	\$350	393	364	92.6%	6	0	23	\$6,055	\$5,102	\$350	36.9%	53.0%
941bd139	PATH	\$400	393	362	92.1%	3	0	28	\$8,189	\$5,584	\$399	39.0%	53.2%
40aae8a2	RET	\$250	291	274	94.2%	17	0	0	\$5,764	\$2,386	\$250	7.9%	7.8%
40aae8a2	RET	\$300	291	284	97.6%	7	0	0	\$7,616	\$3,532	\$300	23.2%	29.7%
40aae8a2	RET	\$350	291	283	97.3%	3	0	5	\$8,848	\$4,611	\$350	28.5%	53.4%
40aae8a2	RET	\$400	291	278	95.5%	0	0	13	\$8,195	\$5,512	\$400	35.7%	55.5%
a6afc475	PERS	\$250	334	328	98.2%	2	0	4	\$8,818	\$5,316	\$250	18.0%	63.0%
a6afc475	PERS	\$300	334	329	98.5%	0	0	5	\$9,286	\$6,472	\$300	18.3%	69.7%
a6afc475	PERS	\$350	334	312	93.4%	0	0	22	\$13,254	\$6,952	\$350	20.9%	66.6%
a6afc475	PERS	\$400	334	307	91.9%	0	0	27	\$14,724	\$8,064	\$400	20.0%	71.2%
744a501f	OPEN	\$250	208	190	91.3%	18	0	0	\$4,338	\$2,023	\$249	10.1%	4.3%
744a501f	OPEN	\$300	208	204	98.1%	4	0	0	\$5,692	\$2,502	\$299	4.7%	8.5%
744a501f	OPEN	\$350	208	207	99.5%	1	0	0	\$6,059	\$3,300	\$350	18.9%	9.9%
744a501f	OPEN	\$400	208	208	100.0%	0	0	0	\$7,400	\$3,803	\$400	33.2%	41.5%
db0a9135	XMKT	\$250	86	86	100.0%	0	0	0	\$3,596	\$2,204	\$248	15.7%	2.7%
db0a9135	XMKT	\$300	86	86	100.0%	0	0	0	\$4,686	\$2,619	\$300	19.1%	15.5%
db0a9135	XMKT	\$350	86	81	94.2%	0	0	5	\$4,635	\$2,706	\$350	18.3%	8.7%
db0a9135	XMKT	\$400	86	79	91.9%	0	0	7	\$6,031	\$3,248	\$399	23.8%	27.9%
d24a71f3	XMKT	\$250	84	70	83.3%	14	0	0	\$2,676	\$1,843	\$249	0.0%	0.0%
d24a71f3	XMKT	\$300	84	80	95.2%	4	0	0	\$1,952	\$2,410	\$300	0.0%	6.2%
d24a71f3	XMKT	\$350	84	82	97.6%	2	0	0	\$5,392	\$3,084	\$350	7.3%	16.5%
d24a71f3	XMKT	\$400	84	83	98.8%	1	0	0	\$6,762	\$3,467	\$393	20.9%	16.9%
ba1e53e9	RET	\$250	99	96	97.0%	0	0	3	\$698	\$2,624	\$250	0.0%	13.1%
ba1e53e9	RET	\$300	99	92	92.9%	0	0	7	\$1,094	\$3,156	\$300	0.3%	15.4%
ba1e53e9	RET	\$350	99	82	82.8%	0	0	17	\$1,872	\$3,136	\$350	4.9%	26.1%
ba1e53e9	RET	\$400	99	80	80.8%	0	0	19	\$3,486	\$3,692	\$400	12.7%	32.5%
39346c8a	TECH	\$250	185	168	90.8%	17	0	0	\$734	\$2,038	\$250	0.0%	0.0%

ID	Fam	Budget	Signals	Exec	Coverage	1-MNQ	Open	Daily	Net MNQ \$	Max DD \$	Max risk \$	Pass	Fail
39346c8a	TECH	\$300	185	179	96.8%	6	0	0	\$1,799	\$2,552	\$298	0.0%	9.9%
39346c8a	TECH	\$350	185	181	97.8%	3	0	1	\$1,796	\$3,212	\$350	1.7%	28.2%
39346c8a	TECH	\$400	185	178	96.2%	2	0	5	\$3,086	\$3,050	\$399	6.4%	30.5%
e61bf491	VOL	\$250	94	81	86.2%	12	0	1	\$1,275	\$1,466	\$248	0.0%	0.0%
e61bf491	VOL	\$300	94	87	92.6%	6	0	1	\$2,077	\$1,658	\$298	0.0%	0.0%
e61bf491	VOL	\$350	94	91	96.8%	2	0	1	\$3,214	\$2,353	\$348	6.2%	11.4%
e61bf491	VOL	\$400	94	91	96.8%	2	0	1	\$3,299	\$2,648	\$399	6.2%	12.0%
98cb83b1	RET	\$250	11	11	100.0%	0	0	0	\$184	\$738	\$248	0.0%	0.0%
98cb83b1	RET	\$300	11	11	100.0%	0	0	0	\$272	\$973	\$300	0.0%	0.0%
98cb83b1	RET	\$350	11	11	100.0%	0	0	0	\$187	\$1,125	\$350	0.0%	0.0%
98cb83b1	RET	\$400	11	11	100.0%	0	0	0	\$371	\$1,264	\$396	0.0%	0.0%
47c1bbe9	XMKT	\$250	107	105	98.1%	2	0	0	\$1,310	\$926	\$248	0.0%	0.0%
47c1bbe9	XMKT	\$300	107	106	99.1%	1	0	0	\$2,116	\$1,328	\$300	0.0%	0.0%
47c1bbe9	XMKT	\$350	107	107	100.0%	0	0	0	\$2,184	\$1,674	\$346	0.5%	0.0%
47c1bbe9	XMKT	\$400	107	107	100.0%	0	0	0	\$2,889	\$1,898	\$400	0.0%	12.2%
9757cb9e	DIST	\$250	639	558	87.3%	73	0	8	\$3,690	\$6,126	\$250	7.3%	58.8%
9757cb9e	DIST	\$300	639	595	93.1%	27	0	17	\$3,369	\$7,803	\$300	3.2%	81.7%
9757cb9e	DIST	\$350	639	574	89.8%	16	0	49	\$-1,623	\$10,453	\$350	10.8%	88.7%
9757cb9e	DIST	\$400	639	559	87.5%	6	0	74	\$2,169	\$8,563	\$400	13.1%	82.0%
1dbad4cd	TECH	\$250	622	584	93.9%	38	0	0	\$3,361	\$6,711	\$250	11.7%	43.9%
1dbad4cd	TECH	\$300	622	607	97.6%	13	0	2	\$4,732	\$8,716	\$300	23.8%	57.0%
1dbad4cd	TECH	\$350	622	594	95.5%	7	0	21	\$4,331	\$9,507	\$350	15.1%	83.2%
1dbad4cd	TECH	\$400	622	581	93.4%	4	0	37	\$9,592	\$8,047	\$400	18.8%	79.4%
b6a51de6	RET	\$250	167	158	94.6%	9	0	0	\$2,542	\$2,324	\$250	0.0%	9.3%
b6a51de6	RET	\$300	167	164	98.2%	3	0	0	\$2,475	\$2,818	\$300	0.0%	14.0%
b6a51de6	RET	\$350	167	164	98.2%	2	0	1	\$1,914	\$3,566	\$348	1.2%	34.1%
b6a51de6	RET	\$400	167	162	97.0%	2	0	3	\$3,155	\$3,948	\$400	6.6%	45.4%
ec410944	OPEN	\$250	208	208	100.0%	0	0	0	\$6,129	\$2,416	\$250	32.3%	4.7%
ec410944	OPEN	\$300	208	208	100.0%	0	0	0	\$6,964	\$3,068	\$300	38.3%	15.7%
ec410944	OPEN	\$350	208	208	100.0%	0	0	0	\$7,246	\$3,622	\$350	39.5%	24.1%
ec410944	OPEN	\$400	208	208	100.0%	0	0	0	\$8,978	\$4,520	\$400	44.8%	25.9%
14478681	RET	\$250	196	180	91.8%	16	0	0	\$2,772	\$1,821	\$249	0.0%	0.0%
14478681	RET	\$300	196	188	95.9%	8	0	0	\$3,504	\$2,226	\$300	0.9%	6.1%
14478681	RET	\$350	196	192	98.0%	4	0	0	\$4,040	\$2,496	\$350	4.0%	9.1%
14478681	RET	\$400	196	194	99.0%	1	0	1	\$5,666	\$3,006	\$400	17.5%	14.3%

ID	Fam	Budget	Signals	Exec	Coverage	1-MNQ	Open	Daily	Net MNQ \$	Max DD \$	Max risk \$	Pass	Fail
158e6884	PATH	\$250	95	89	93.7%	6	0	0	\$1,886	\$706	\$250	0.0%	0.0%
158e6884	PATH	\$300	95	90	94.7%	5	0	0	\$2,543	\$891	\$300	0.0%	0.0%
158e6884	PATH	\$350	95	94	98.9%	1	0	0	\$1,995	\$1,739	\$350	0.0%	0.0%
158e6884	PATH	\$400	95	94	98.9%	1	0	0	\$3,212	\$1,416	\$400	0.0%	0.0%
d24243de	RET	\$250	215	185	86.0%	30	0	0	\$6,504	\$1,148	\$250	0.5%	0.0%
d24243de	RET	\$300	215	196	91.2%	19	0	0	\$7,119	\$1,485	\$300	9.9%	0.0%
d24243de	RET	\$350	215	206	95.8%	6	0	3	\$7,577	\$1,896	\$350	18.6%	0.0%
d24243de	RET	\$400	215	209	97.2%	3	0	3	\$9,788	\$2,386	\$398	25.6%	22.9%
969b7274	REOPEN	\$250	409	318	77.8%	91	0	0	\$-4,348	\$7,327	\$250	0.0%	48.3%
969b7274	REOPEN	\$300	409	350	85.6%	59	0	0	\$-3,525	\$8,936	\$300	3.4%	60.5%
969b7274	REOPEN	\$350	409	375	91.7%	32	0	2	\$-7,369	\$14,154	\$350	3.4%	89.2%
969b7274	REOPEN	\$400	409	392	95.8%	15	0	2	\$-3,724	\$13,014	\$400	3.5%	91.6%
60857891	VWAP	\$250	199	198	99.5%	1	0	0	\$5,182	\$3,346	\$250	3.2%	9.6%
60857891	VWAP	\$300	199	197	99.0%	1	0	1	\$6,736	\$4,096	\$300	8.4%	16.2%
60857891	VWAP	\$350	199	192	96.5%	0	0	7	\$8,988	\$3,694	\$350	14.5%	18.3%
60857891	VWAP	\$400	199	190	95.5%	0	0	9	\$10,622	\$4,991	\$399	25.2%	22.7%
efad812f	TECH	\$250	227	215	94.7%	12	0	0	\$1,668	\$1,336	\$249	0.0%	0.0%
efad812f	TECH	\$300	227	221	97.4%	6	0	0	\$1,189	\$1,998	\$300	0.0%	7.0%
efad812f	TECH	\$350	227	223	98.2%	3	0	1	\$1,398	\$2,117	\$350	0.0%	14.3%
efad812f	TECH	\$400	227	222	97.8%	2	0	3	\$2,416	\$2,406	\$400	0.0%	17.5%

2026 forward

ID	Fam	Budget	Signals	Exec	Coverage	1-MNQ	Open	Daily	Net MNQ \$	Max DD \$	Max risk \$	Pass	Fail
941bd139	PATH	\$250	71	34	47.9%	37	0	0	\$1,962	\$760	\$250	0.0%	0.0%
941bd139	PATH	\$300	71	45	63.4%	26	0	0	\$3,704	\$589	\$297	0.0%	0.0%
941bd139	PATH	\$350	71	52	73.2%	19	0	0	\$5,618	\$748	\$348	100.0%	0.0%
941bd139	PATH	\$400	71	59	83.1%	10	0	2	\$6,110	\$1,056	\$396	100.0%	0.0%
40aae8a2	RET	\$250	74	43	58.1%	31	0	0	\$668	\$902	\$249	0.0%	0.0%
40aae8a2	RET	\$300	74	54	73.0%	20	0	0	\$3,192	\$644	\$299	63.2%	0.0%
40aae8a2	RET	\$350	74	63	85.1%	10	0	1	\$2,422	\$1,011	\$350	78.9%	0.0%
40aae8a2	RET	\$400	74	69	93.2%	4	0	1	\$4,628	\$1,066	\$396	89.5%	0.0%
a6afc475	PERS	\$250	63	60	95.2%	1	0	2	\$2,506	\$2,032	\$248	0.0%	97.4%
a6afc475	PERS	\$300	63	61	96.8%	0	0	2	\$2,115	\$2,510	\$297	0.0%	100.0%
a6afc475	PERS	\$350	63	58	92.1%	0	0	5	\$3,904	\$3,432	\$350	2.6%	92.1%
a6afc475	PERS	\$400	63	55	87.3%	0	0	8	\$5,303	\$3,989	\$396	10.5%	89.5%
744a501f	OPEN	\$250	50	9	18.0%	41	0	0	\$-128	\$678	\$242	0.0%	0.0%

ID	Fam	Budget	Signals	Exec	Coverage	1-MNQ	Open	Daily	Net MNQ \$	Max DD \$	Max risk \$	Pass	Fail
744a501f	OPEN	\$300	50	24	48.0%	26	0	0	\$1,538	\$786	\$292	0.0%	0.0%
744a501f	OPEN	\$350	50	42	84.0%	8	0	0	\$2,154	\$1,920	\$347	0.0%	28.9%
744a501f	OPEN	\$400	50	48	96.0%	2	0	0	\$2,996	\$1,176	\$399	0.0%	0.0%
db0a9135	XMKT	\$250	9	9	100.0%	0	0	0	\$806	\$429	\$247	0.0%	0.0%
db0a9135	XMKT	\$300	9	9	100.0%	0	0	0	\$806	\$429	\$247	0.0%	0.0%
db0a9135	XMKT	\$350	9	9	100.0%	0	0	0	\$1,216	\$617	\$317	0.0%	0.0%
db0a9135	XMKT	\$400	9	9	100.0%	0	0	0	\$1,682	\$738	\$392	0.0%	0.0%
d24a71f3	XMKT	\$250	15	2	13.3%	13	0	0	\$-8	\$242	\$246	0.0%	0.0%
d24a71f3	XMKT	\$300	15	5	33.3%	10	0	0	\$437	\$242	\$284	0.0%	0.0%
d24a71f3	XMKT	\$350	15	8	53.3%	7	0	0	\$1,714	\$242	\$340	0.0%	0.0%
d24a71f3	XMKT	\$400	15	13	86.7%	2	0	0	\$2,114	\$394	\$392	0.0%	0.0%
ba1e53e9	RET	\$250	30	30	100.0%	0	0	0	\$2,390	\$705	\$248	18.4%	0.0%
ba1e53e9	RET	\$300	30	30	100.0%	0	0	0	\$3,952	\$840	\$298	100.0%	0.0%
ba1e53e9	RET	\$350	30	27	90.0%	0	0	3	\$3,675	\$1,080	\$350	100.0%	0.0%
ba1e53e9	RET	\$400	30	26	86.7%	0	0	4	\$4,408	\$1,122	\$396	100.0%	0.0%
39346c8a	TECH	\$250	28	15	53.6%	13	0	0	\$460	\$540	\$246	0.0%	0.0%
39346c8a	TECH	\$300	28	18	64.3%	10	0	0	\$48	\$802	\$296	0.0%	0.0%
39346c8a	TECH	\$350	28	25	89.3%	3	0	0	\$1,672	\$1,208	\$346	0.0%	0.0%
39346c8a	TECH	\$400	28	26	92.9%	2	0	0	\$1,488	\$1,408	\$395	0.0%	0.0%
e61bf491	VOL	\$250	26	14	53.8%	12	0	0	\$644	\$224	\$246	0.0%	0.0%
e61bf491	VOL	\$300	26	20	76.9%	6	0	0	\$2,896	\$224	\$298	0.0%	0.0%
e61bf491	VOL	\$350	26	20	76.9%	6	0	0	\$2,980	\$323	\$343	0.0%	0.0%
e61bf491	VOL	\$400	26	23	88.5%	3	0	0	\$2,010	\$496	\$393	0.0%	0.0%
98cb83b1	RET	\$250	2	2	100.0%	0	0	0	\$42	\$22	\$162	0.0%	0.0%
98cb83b1	RET	\$300	2	2	100.0%	0	0	0	\$106	\$22	\$299	0.0%	0.0%
98cb83b1	RET	\$350	2	2	100.0%	0	0	0	\$83	\$45	\$325	0.0%	0.0%
98cb83b1	RET	\$400	2	2	100.0%	0	0	0	\$83	\$45	\$325	0.0%	0.0%
47c1bbe9	XMKT	\$250	23	2	8.7%	21	0	0	\$29	\$226	\$224	0.0%	0.0%
47c1bbe9	XMKT	\$300	23	11	47.8%	12	0	0	\$2,088	\$524	\$296	0.0%	0.0%
47c1bbe9	XMKT	\$350	23	15	65.2%	8	0	0	\$3,114	\$298	\$348	0.0%	0.0%
47c1bbe9	XMKT	\$400	23	20	87.0%	3	0	0	\$5,098	\$436	\$382	78.9%	0.0%
9757cb9e	DIST	\$250	131	63	48.1%	67	0	1	\$-2,323	\$2,528	\$250	0.0%	36.8%
9757cb9e	DIST	\$300	131	87	66.4%	43	0	1	\$-1,999	\$2,779	\$298	0.0%	100.0%
9757cb9e	DIST	\$350	131	102	77.9%	25	0	4	\$-513	\$2,258	\$349	0.0%	100.0%
9757cb9e	DIST	\$400	131	111	84.7%	10	0	10	\$2,092	\$1,904	\$397	0.0%	100.0%

ID	Fam	Budget	Signals	Exec	Coverage	1-MNQ	Open	Daily	Net MNQ \$	Max DD \$	Max risk \$	Pass	Fail
1dbad4cd	TECH	\$250	137	85	62.0%	52	0	0	\$1,528	\$1,918	\$249	10.5%	0.0%
1dbad4cd	TECH	\$300	137	111	81.0%	26	0	0	\$1,602	\$2,454	\$300	10.5%	89.5%
1dbad4cd	TECH	\$350	137	116	84.7%	17	0	4	\$2,510	\$2,344	\$350	78.9%	21.1%
1dbad4cd	TECH	\$400	137	121	88.3%	9	0	7	\$3,796	\$2,932	\$400	60.5%	39.5%
b6a51de6	RET	\$250	42	29	69.0%	13	0	0	\$-528	\$796	\$245	0.0%	0.0%
b6a51de6	RET	\$300	42	33	78.6%	9	0	0	\$223	\$899	\$296	0.0%	0.0%
b6a51de6	RET	\$350	42	37	88.1%	5	0	0	\$-431	\$1,348	\$349	0.0%	0.0%
b6a51de6	RET	\$400	42	39	92.9%	3	0	0	\$57	\$1,388	\$398	0.0%	0.0%
ec410944	OPEN	\$250	50	50	100.0%	0	0	0	\$2,300	\$601	\$242	0.0%	0.0%
ec410944	OPEN	\$300	50	50	100.0%	0	0	0	\$3,792	\$541	\$300	89.5%	0.0%
ec410944	OPEN	\$350	50	50	100.0%	0	0	0	\$3,717	\$1,389	\$347	89.5%	0.0%
ec410944	OPEN	\$400	50	50	100.0%	0	0	0	\$4,334	\$1,278	\$400	100.0%	0.0%
14478681	RET	\$250	48	38	79.2%	9	0	1	\$-101	\$864	\$249	0.0%	0.0%
14478681	RET	\$300	48	39	81.2%	8	0	1	\$-415	\$1,178	\$298	0.0%	0.0%
14478681	RET	\$350	48	41	85.4%	6	0	1	\$-134	\$1,150	\$344	0.0%	0.0%
14478681	RET	\$400	48	43	89.6%	4	0	1	\$-685	\$1,590	\$400	0.0%	0.0%
158e6884	PATH	\$250	18	9	50.0%	9	0	0	\$-332	\$566	\$240	0.0%	0.0%
158e6884	PATH	\$300	18	13	72.2%	5	0	0	\$397	\$566	\$286	0.0%	0.0%
158e6884	PATH	\$350	18	15	83.3%	3	0	0	\$588	\$566	\$342	0.0%	0.0%
158e6884	PATH	\$400	18	16	88.9%	2	0	0	\$-92	\$902	\$391	0.0%	0.0%
d24243de	RET	\$250	53	23	43.4%	30	0	0	\$-1,028	\$1,373	\$239	0.0%	0.0%
d24243de	RET	\$300	53	31	58.5%	22	0	0	\$446	\$1,216	\$298	0.0%	0.0%
d24243de	RET	\$350	53	40	75.5%	13	0	0	\$-102	\$1,844	\$348	0.0%	0.0%
d24243de	RET	\$400	53	48	90.6%	5	0	0	\$976	\$2,254	\$398	10.5%	10.5%
969b7274	REOPEN	\$250	82	12	14.6%	70	0	0	\$-114	\$588	\$240	0.0%	0.0%
969b7274	REOPEN	\$300	82	24	29.3%	58	0	0	\$-526	\$702	\$300	0.0%	0.0%
969b7274	REOPEN	\$350	82	38	46.3%	44	0	0	\$-1,692	\$1,754	\$346	0.0%	0.0%
969b7274	REOPEN	\$400	82	49	59.8%	33	0	0	\$-2,176	\$2,696	\$399	0.0%	100.0%
60857891	VWAP	\$250	50	42	84.0%	8	0	0	\$-1,086	\$1,238	\$242	0.0%	0.0%
60857891	VWAP	\$300	50	47	94.0%	3	0	0	\$-641	\$1,314	\$294	0.0%	0.0%
60857891	VWAP	\$350	50	46	92.0%	2	0	2	\$-740	\$1,430	\$344	0.0%	0.0%
60857891	VWAP	\$400	50	47	94.0%	1	0	2	\$-428	\$1,760	\$398	0.0%	0.0%
efad812f	TECH	\$250	44	23	52.3%	21	0	0	\$-900	\$1,044	\$248	0.0%	0.0%
efad812f	TECH	\$300	44	33	75.0%	11	0	0	\$-863	\$1,207	\$296	0.0%	0.0%
efad812f	TECH	\$350	44	40	90.9%	4	0	0	\$-1,068	\$1,412	\$346	0.0%	0.0%

ID	Fam	Budget	Signals	Exec	Coverage	1-MNQ	Open	Daily	Net MNQ \$	Max DD \$	Max risk \$	Pass	Fail
efad812f	TECH	\$400	44	43	97.7%	1	0	0	\$-695	\$1,384	\$395	0.0%	0.0%

E. Prop-evaluation tables

OFFICIAL FIVE + POST-HOC BUDGETS

Section 10 reproduces all saved \$250-\$400 all-survivor combination evaluation rows, including pass and fail counts and rates, unresolved rates, median days, net MNQ P&L, drawdown, simultaneous risk and signal coverage. Below are the original-five budget sensitivities, including the saved 90th-percentile days to pass.

Period	Budget	Starts	Pass	Fail	Unresolved	Median days	P90 days	EOD net MNQ \$	Max DD \$
2023-25	\$200	656	298 / 45.4%	322 / 49.1%	5.5%	75.0	92.0	\$10,761	\$6,464
2023-25	\$250	656	212 / 32.3%	386 / 58.8%	8.8%	66.5	96.9	\$10,606	\$8,490
2023-25	\$300	656	214 / 32.6%	406 / 61.9%	5.5%	53.0	82.0	\$16,026	\$10,087
2023-25	\$350	656	221 / 33.7%	423 / 64.5%	1.8%	51.0	87.0	\$17,480	\$8,904
2023-25	\$400	656	284 / 43.3%	372 / 56.7%	0.0%	39.0	89.7	\$26,939	\$9,348
2026	\$200	38	0 / 0.0%	27 / 71.1%	28.9%	n/a	n/a	\$-2,724	\$2,724
2026	\$250	38	0 / 0.0%	38 / 100.0%	0.0%	n/a	n/a	\$-2,920	\$3,724
2026	\$300	38	0 / 0.0%	0 / 0.0%	100.0%	n/a	n/a	\$363	\$1,805
2026	\$350	38	36 / 94.7%	2 / 5.3%	0.0%	49.5	67.0	\$3,120	\$2,068
2026	\$400	38	5 / 13.2%	33 / 86.8%	0.0%	29.0	30.8	\$9,300	\$2,210

All percentages are in-sample rolling-start descriptions, not independent estimates of future pass probability. EOD and intraday profiles are both saved in source JSON. The original five at \$200 are the frozen policy; other budgets remain post-hoc.

F. Null-world distributions

OFFICIAL DISCOVERY + POST-HOC EXTENSION			
Post-hoc pipeline metric	Real	A median / 95th / max	C median / 95th / max
Validation	43	4 / 12.0 / 18	3 / 13.0 / 16
Holdout supported	16	0 / 2.0 / 4	0 / 2.0 / 4
2026 supported	13	0 / 3.0 / 5	0 / 3.0 / 6
Both supported	4	0 / 1.0 / 1	0 / 1.0 / 2
Distinct both	3	0 / 1.0 / 1	0 / 1.0 / 2

Official discovery nulls used 150 A, 150 C and 40 B worlds. Post-hoc later-survivor nulls used 100 A and 100 C worlds, with separately generated later null markets. Neither the later-survivor tails nor combined A/C counts are new preregistered p-values.

G. Methodology and freeze timeline

OFFICIAL FREEZE RECORD

Freeze body SHA-256 hashes

Freeze	Body SHA-256
Prereg	b524704309412a65c879d45e85efb439ed703ac586b748fd55d0dc6cc54f0ac9
Discovery	e10e0c0d456bfae76ae482bdf839f044e3307ac53b56bb30abbaf73a30fbe0ff
Confirmation	fb67e008ca5d859f24efd101b8b25d90e401941093e96c9c4a741831b7084a9e
Final cohort	1708eddcf919a46daef57cf8b01014f8e196f4e39f20acb08842fb2604b7f307
Holdout report	470eda93ea19e834f4d122a716989ef897cc609ccb87fd1d55658d7df3d572d4

The final cohort froze at 2026-09-25T05:20:01Z before the 2026 sealed audit. Key git tags: v4-prereg, v4-discovery-freeze, v4-validation-freeze, v4-final-cohort-freeze, v4-holdout-sealed, v4-campaign1-complete. The report builder reads but does not modify any freeze or source result.

H. Git commits and source checksums

SOURCE INTEGRITY	
Git tag	Commit SHA
v4-prereg	532a87818d57545d7fc6af13733ac21759bbcc34
v4-discovery-freeze	9c6c68354e3a8141ff653335f80c6e0eefcd4693
v4-validation-freeze	00e2399666b3645d47a8118fd75c78a758cbb49d
v4-final-cohort-freeze	b3dc94e9ebc7d75a4eb462f3ec4c91e1d2e6026a
v4-holdout-sealed	4c8f5f85e65388dfa09df5e4b03821e5d68c1fd1
v4-campaign1-complete	f491c796b85920c538ea05e18a2b9644d638ac24

Source-file checksums (SHA-256)

Source	SHA-256
V4_MASTER_RESEARCH_RECORD.md	81123a4973814f0507d84a12a6ac31b114dacea61d378ae85918688d8ebe8cf6
V4_DISCOVERY_NULL_REPORT.md	139697728f7488811d598e584ccd2d0fe831270623d3d68294b15c5b2d4380cb
V4_POSTHOC_ALL_VALIDATION_SURVIVORS.json	ad5f5461ba9d70afb5b8fa0b83899d7d41b645bd93f23ebb99c5a8609d84f7c9
V4_POSTHOC_NULL_SURVIVOR_EXTENSION.json	57d8981da886bdf8fb5e72b711469bbcab9ef47a07f17395b8b52ba1b90a290
results/posthoc/ALL_VALIDATION_SURVIVORS_TRADES.npz	d16a3366e458f87b46a3538e365481ec06aa16b5dde604301ffff83d1d060610
results/posthoc/RISK_BUDGET_SENSITIVITY.json	a00363acc145888b56c1bc670cf8328019a5ccb33df63ff005e0611ae8a6ee02
results/posthoc/RISK_CAP_DIAGNOSTIC.json	c56d460f3c4a9f756beb86e14674c43154b96bd97afdc8cc19812fec0ba6b0d7

I. Limitations and post-hoc disclosures

POST-HOC DISCLOSURE

1. The official top-five and \$200 policy remain unchanged. All-43 testing, dual support, portfolios and risk budgets are post-hoc after later outcomes were known.
2. Only 100 A and 100 C worlds were used in the all-survivor extension; plus-one p cannot resolve below 0.0099 within either kind. A/C are related and are not 200 independent trials.
3. Directional sign flips remove market drift; null validation cohorts are much smaller than the real 43. Null B preserves 30-minute paths and gives weaker official discovery evidence.
4. No family survives Holm correction. Some planted small edges were missed at original strength; V4 has limited power near its search maximum.
5. Individual discovery-period full trade/profile metrics for the four IDs were not present in the report's accessible saved survivor bundle; discovery cells are marked unavailable. The exact trigger codes are preserved without inventing missing formulas.
6. The saved prop summary does not expose 90th percentile time to pass, worst week or a full intraday drawdown trace for every configuration. Averages and curves are shown only where exact saved trade streams allow them.
7. The 2026 period ends 10 August 2026; it is not a full calendar year. Thirty-eight rolling 120-session starts overlap heavily. Earlier V2/V3 exposure limits researcher blinding even though V4 stage access was frozen.
8. MNQ commission is an unvalidated \$1 round-trip placeholder. Generic prop rules require firm-specific verification. Live-forward observations were not available in the campaign record.

Report generated from existing project results. Source: output/pdf/build_v4_complete_report.py. Any future V5 experiment should be newly preregistered and judged on data not used to diagnose V4.